



Sec: Sr. IIT

Race -18

Date: 04-03-24

PHYSICS

Max Marks: 62

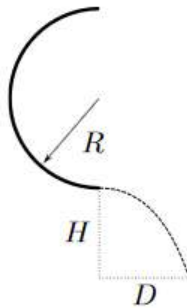
SECTION – I

(SINGLE CORRECT ANSWER TYPE)

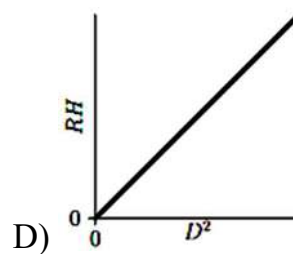
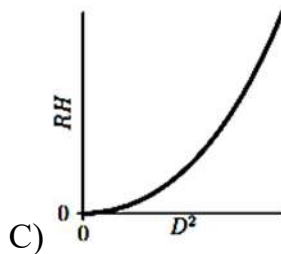
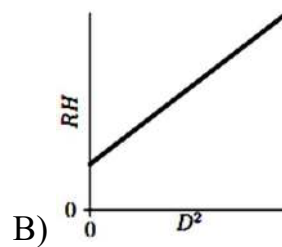
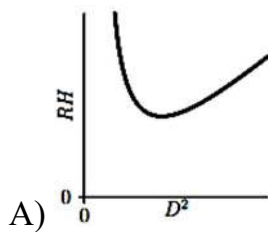
This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

1. A semicircular wire of radius R is oriented vertically. A small bead is released from rest at the top of the wire, it slides without friction under the influence of gravity to the bottom, where it then leaves the wire horizontally and falls down a distance H to the ground. The bead lands a horizontal distance D away from where it was launched



Which of the following is a correct graph of RH against D^2 ?



2. An open parachute of mass 1.0 kg is coming straight down from the sky. Attached to the parachute is the upper end of a light spring scale, while a block of mass 10 kg is attached to its lower end of the scale. The scale reading is 80 N. The air resistance at the moment is approximately.
A) 55 N B) 66 N C) 77 N D) 88 N
3. A uniform thin circular rubber band of mass M and spring constant k has an original radius R . Now it is tossed into the air. Assume it remains circular when stabilized in air and rotates at angular speed ω about its centre uniformly. Which of the following gives the new radius of the rubber band?
A) $(2\pi kR)/(2\pi k - M\omega^2)$ B) $(4\pi kR)/(4\pi k - M\omega^2)$
C) $(8\pi^2 kR)/(8\pi^2 k - M\omega^2)$ D) $(4\pi^2 kR)/(4\pi^2 k - M\omega^2)$
4. A spherical cloud of dust in space has a uniform density ρ_0 and a radius R_0 . The gravitational acceleration of free fall at the surface of the cloud due to the mass of the cloud is g_0 . A process occurs (heat expansion) that causes the cloud to suddenly grow to a radius $2R_0$, while maintaining a uniform (but not constant) density. The gravitational acceleration of free fall at a point R_0 away from the centre of the cloud due to the mass of the cloud is now
A) $g_0/32$ B) $g_0/16$ C) $g_0/8$ D) $g_0/4$
5. Three students make measurements of the length of a 1.50 meter rod. Each student reports an uncertainty estimate representing an independent random error applicable to the measurement.
Alice: A single measurement using a 2.0 meter long tape measure, to within ± 2 mm.
Bob: Two measurements using a wooden meter stick, each to within ± 2 mm, which he adds together.
Christina: Two measurements using a machinist's meter ruler, each to within ± 1 mm, which she adds together.
The students' teacher prefers measurements that are likely to have less percentage relative error. Which is the correct order of preference?
A) Christina's is preferable to Alice's, which is preferable to Bob's
B) Alice's is preferable to Christina's, which is preferable to Bob's
C) Alice's and Christina's are equally preferable; both are preferable to Bob's
D) Christina's is preferable to both Alice's and to Bob's, which are equally preferable

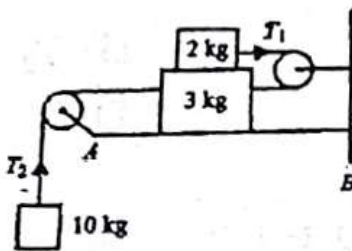
SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

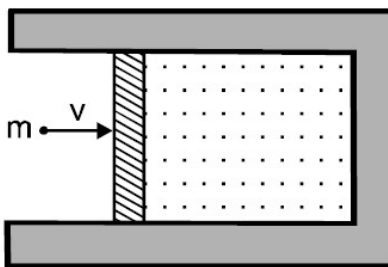
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

6. An object of mass m_1 initially moving at speed v_0 collides with an originally stationary object of mass $m_2 = \alpha m_1$, where $\alpha < 1$. The collision could be completely elastic, completely inelastic, or partially inelastic. After the collision the two objects move at speeds v_1 and v_2 . Assume that the collision is one dimensional, and that object one cannot pass through object two. After the collision, the speed ratio $r_1 = v_1/v_0$ and $r_2 = v_2/v_0$ are bounded by
- A) $(1 - \alpha)/(1 + \alpha) \leq r_1 \leq 1$ B) $(1 - \alpha)/(1 + \alpha) \leq r_1 \leq 1/(1 + \alpha)$
 C) $\alpha/(1 + \alpha) \leq r_2 \leq 1$ D) $1/(1 + \alpha) \leq r_2 \leq 2/(1 + \alpha)$
7. Accelerating charges radiate. The total power P radiated by charge q with acceleration a is given by $P = C \xi a^n$ where C is a dimensionless numerical constant (which is equal to $\frac{1}{6\pi}$), ξ is a physical constant that is a function only of the charge q , the speed of light c , and the permittivity of free space ϵ_0 , and n is a dimensionless constant. Then which of the following are correct
- A) if acceleration a becomes 2 times then power radiated becomes 4 times
 B) if acceleration a becomes 2 times then power radiated becomes 8 times
 C) if charge q becomes 2 times then power radiated becomes 4 times
 D) if charge q becomes 2 times then power radiated becomes 8 times
8. Coefficient of friction between the two blocks is 0.3. Whereas the surface AB is smooth



- A) Acceleration of the system of masses is $88/15 \text{ ms}^{-2}$
 B) Net force acting on 3 kg mass is greater than that on 2 kg mass
 C) Tension $T_2 > T_1$
 D) Since 10 kg mass is accelerating downwards, so net force acting on it should be greater than any of the two blocks shown in the figure

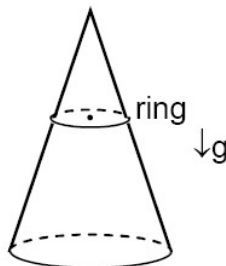
9. In a flexible balloon, 2 moles of SO_2 having initial volume of 1m^3 at a temperature of 27°C is filled (SO_2 is a non-linear triatomic gas). The gas is first expanded to thrice its initial volume isobarically and then expanded adiabatically so as to attain its initial temperature. Assuming gas is ideal and $R=25/3 \text{ Jmol}^{-1}\text{K}^{-1}$. Choose the CORRECT option(s).
- A) Change in internal energy of the gas in the isobaric process is $2.5 \times 10^4 \text{ J}$
- B) Change in internal energy of the gas in the isobaric process is $3 \times 10^4 \text{ J}$
- C) Work done by the gas in the whole process is 35 kJ
- D) Work done by the gas in the whole process is 40 kJ
- 10 Consider one mole of an ideal monoatomic gas inside a fixed, perfectly insulating cylindrical vessel at temperature T and piston is in equilibrium initially as shown in figure. Now a particle of mass m moving with velocity v strikes horizontally to piston of same mass at midpoint of piston and sticks to piston. Initially



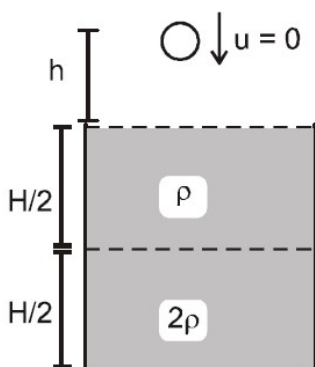
temperature $T = mv^2 / R$, where R is universal gas constant (assume the piston is perfect adiabatic and also velocity of the piston is so small that temperature redistribution of gas due to motion of piston is spontaneous throughout the bulk) (Gravity and friction is absent everywhere) Choose the correct option(s)

- A) Ratio of maximum temperature to minimum temperature of gas during motion of piston is $7 : 5$
- B) Ratio of maximum temperature to minimum temperature of gas during motion of piston is $4 : 3$
- C) If particle strikes elastically then ratio of maximum temperature to initial temperature during motion of piston is $4:3$
- D) If particle strikes elastically then ratio of maximum temperature to minimum temperature during motion of piston is $4 : 3$

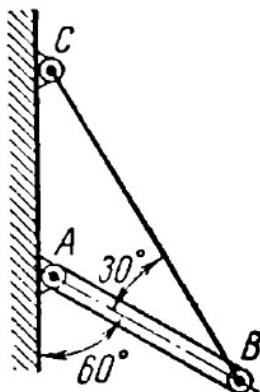
11. An elastic ring of mass 'm' and force constant k (stiffness) is released from rest in a smooth cone of semi vertex angle θ from horizontal position as shown in figure. Initially the ring was in natural length : Choose the incorrect option(s)



- A) ring executes oscillations
 B) initial acceleration of centre of ring is $g \cos^2 \theta$
 C) maximum vertical displacement of centre of ring is $\frac{mg \cot \theta}{2\pi^2 k}$
 D) at the moment of maximum vertical displacement acceleration of centre of ring is zero.
12. A small ball of mass m is released from rest from a height h above the liquid surface. The density of ball is $\rho/3$.
 (assume no splash and area of the liquid surface is very large)



- A) If $h \geq H$, then the ball will be able to enter the liquid of higher density
 B) If $h \geq 7H/2$, then the ball will be able to strike the bottom of the vessel
 C) If $h = 2H$, then the ball will cross the interface of two liquids with kinetic energy $= mgH$
 D) If $h = 0$ then ball will float on the surface
13. A uniform rod AB of mass 2kg is hinged at A to a vertical wall and is held at an angle of 60° to the vertical by a string BC, forming an angle of 30° with the rod (as shown in fig). Then



- A) The magnitude of the reaction force is 10N
- B) The magnitude of the reaction force is 20N
- C) The angle made by reaction force with AC is 45°
- D) The angle made by reaction force with AC is 60°

SECTION – III

(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

14. A 3.0 meter long mass less rod is free to rotate horizontally about its centre. Two 5.0kg point objects are originally located at the ends of the rod; they are free to slide on the frictionless rod and are kept from flying off the rod by an inflexible mass less rope that connects the two objects. Originally the system is rotating at 4.0 radians per second; assume the system is completely frictionless; and ignore any concerns about instability of the system. The rope is slowly tightened by a small mass less motor attached to one of the objects. It is done in such a way as to pull the two objects closer to the centre of the rotating rod. The work is done by the motor in pulling the two objects from the ends of the rod until they are each 0.5meters from the center of rotation is given by ten times of 12^n J then the n is given by
15. The moment of inertia of a uniform equilateral triangle with mass m and side length a about an axis through one of its sides and parallel to that side is $\frac{1}{8}ma^2$. The moment of inertia of a uniform regular hexagon of mass m and side length a about an axis through two opposite vertices is given by $(A/6B)ma^2$. Then A+B
16. The power output from a certain experimental car design to be shaped like a cube is proportional to the mass m of the car. The force of air friction on the car is proportional to Av^2 , where v is the speed of the car and A the cross sectional area. On

a level surface the car has a maximum speed $v_{\max}$. Assuming that all versions of this design have the same density, $v_{\max} \propto m^{1/n}$ then n is given by

17. A gas is enclosed in a vessel of volume V at a pressure of 144 cm of Hg. It is being pumped out of the vessel by means of a piston pump. After two strokes the pressure in the vessel is 36 cm of Hg. Find the pressure in cm of mercury after another two strokes.
18. A satellite is describing a circular orbit around a fixed planet of radius R . The altitude of the satellite above surface of planet is $3R$ and its speed is V_0 . To place the satellite in an elliptical orbit which will bring it closer to the planet, its velocity is reduced from V_0 to βV_0 , when $\beta < 1$. The smallest permissible value of β if satellite is not to crash on the surface of planet is k . Then find k .

CHEMISTRY

Max Marks: 62

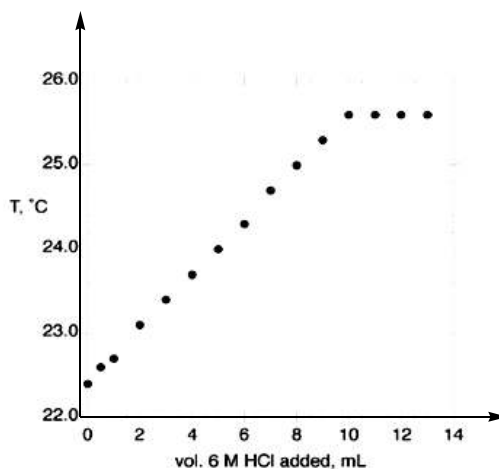
SECTION – I

(SINGLE CORRECT ANSWER TYPE)

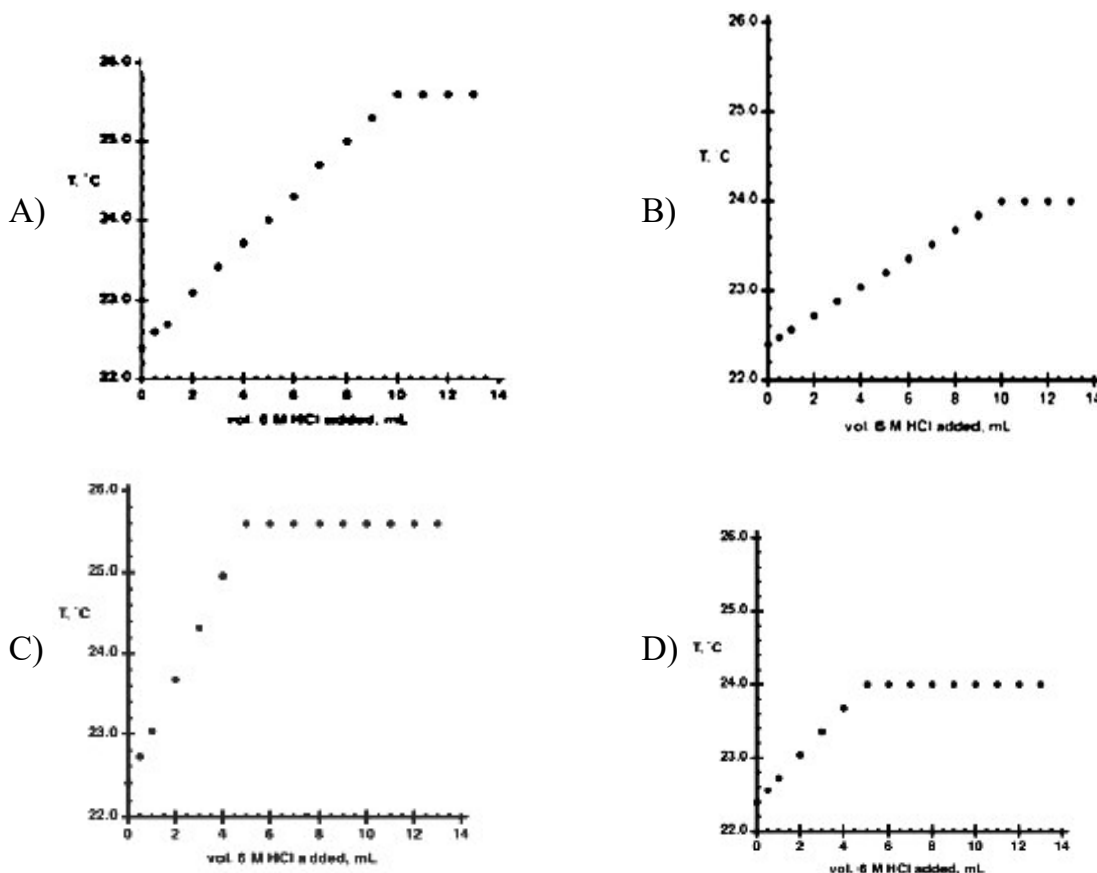
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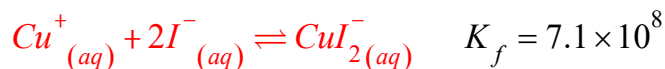
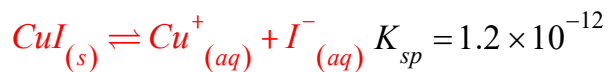
19. A solution containing 0.060 mol NaOH dissolved in 200 g water was treated with successive aliquots of 6.0 M aqueous HCl in a well-insulated flask, and the temperature was measured after each aliquot to give the following data:



The same experiment was repeated, again using 0.060 mol NaOH and 6.0 M aqueous HCl, but this time the NaOH was initially dissolved in 400 g water. Which graph represents the data obtained in this experiment?



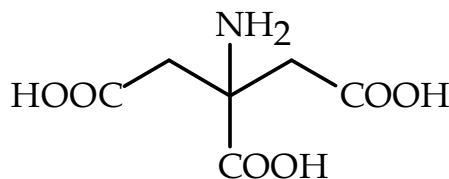
20. Consider these reactions among copper and iodine (all reactions at 298 K):



Calculate the number of moles of copper ions that dissolve if 1.00×10^{-3} mol $\text{CuI}_{(s)}$ is suspended in 1.00 L of solution

- A) 2.1×10^{-8} B) 2.1×10^{-7} C) 1.1×10^{-6} D) 8.2×10^{-5}

21. The correct IUPAC name of the given compound is



- A) 3- Carboxy-3-hydroxy butane dioic acid
 B) 2-Amino propane-1,2,3-tricarboxylic acid
 C) 3-Amino butanedioic acid
 D) 2-Bis(carboxymethyl)-2-hydroxy ethanoic acid
22. Sodium peroxide which is a yellow solid, when exposed to air becomes white due to the formation of
 A) H_2O_2 B) Na_2O C) Na_2O and O_3 D) NaOH and Na_2CO_3
23. A student prepared a sample of silicon chloride by passing chlorine over heated silicon and collecting the condensed chloride by dissolving a known mass of it in water and titrating the solution with standard silver nitrate solution. The formula of the silicon chloride as obtained by this method was $\text{SiCl}_{2.6}$ as against a true formula SiCl_4 . Which of the following possible errors could have resulted in this wrong formula?
 A) The silicon chloride contained excess dissolved chlorine
 B) More silicon chloride than the student supposed was actually used owing to inaccurate weighing
 C) The small specimen tube was not dry
 D) The reaction between the silicon and the chlorine stopped prematurely leaving some unreacted silicon in the reaction tube.

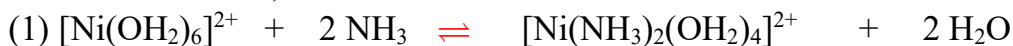
SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

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24. The following two chemical equilibria exist in dilute aqueous solutions at 298 K (en = ethane-1,2-diamine):



for (1) $\ln K_c = 11.60$ and $\Delta_R H^\circ = -33.5 \text{ kJ}$ for (2) $\ln K_c = 17.78$ and $\Delta_R H^\circ = -37.2 \text{ kJ}$

Select the correct statement:

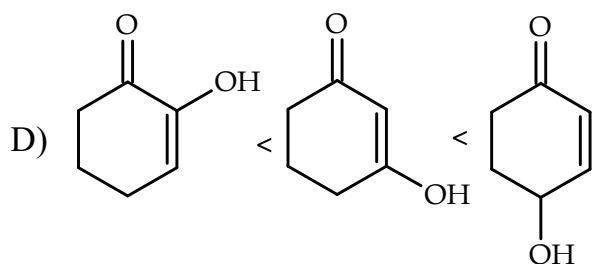
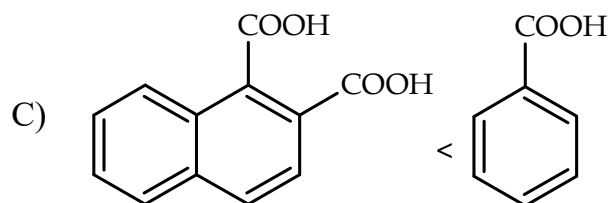
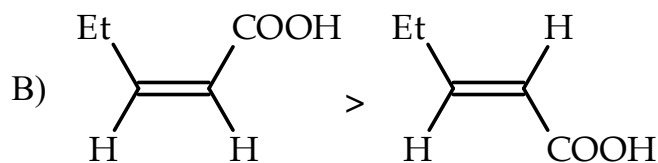
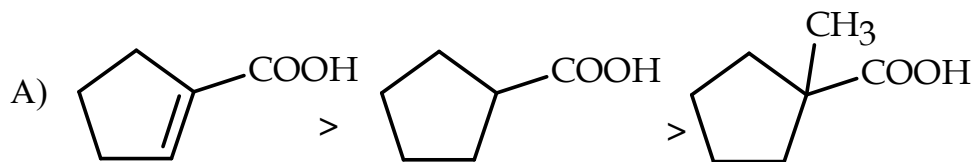
- A) $[\text{Ni}(\text{NH}_3)_2(\text{OH}_2)_4]^{2+}$ is thermodynamically less stable than $[\text{Ni}(\text{en})(\text{OH}_2)_4]^{2+}$
 B) (1) has lower $\Delta_R S^\circ$ than (2)
 C) $\Delta_R H^\circ_{(3)}$ is -3.7 kJ

D) $\Delta_R S^\circ_{(3)}$ is 38.96 JK^{-1}

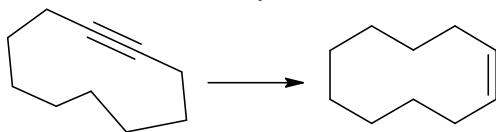
25. A 0.0343 mol sample of $\text{XCl}_{3(g)}$ is added to an evacuated 1.50 L glass bulb. The bulb is sealed, heated to 100°C and the following equilibrium is established:

$2\text{XCl}_{3(g)} \rightleftharpoons 2\text{X}_{(g)} + 3\text{Cl}_{2(g)}$. At equilibrium, the final pressure in the bulb is 0.800 atm. What happens if volume of bulb is doubled?

- A) Partial pressure of XCl_3 is less than 0.4 atm
 B) Partial pressure of X is less than 0.4 atm
 C) Partial pressure of Cl_2 is more than 0.04 atm
 D) Partial pressure of X becomes $2/3$ times that of chlorine at equilibrium
26. Which of the following order is/are correct for acidic strength?

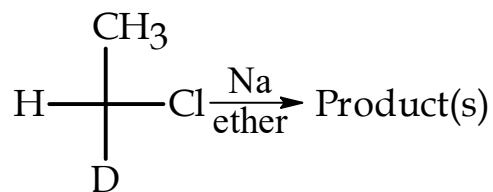


27. The following conversion can be done by

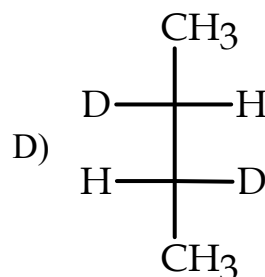
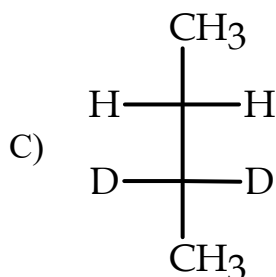
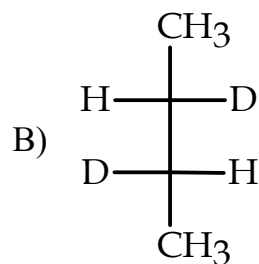
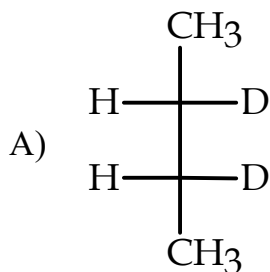


- A) $\text{H}_2/\text{Pd-C}$ B) $\text{H}_2/\text{Ni}_2\text{B}$
 C) $\text{Na}/\text{liq. NH}_3$ D) $\text{H}_2/\text{Pd-BaSO}_4\text{-Quinoline}$

28.



Which product(s) should not form during reaction?



29. Which of the following statements is/are correct?

- A) NaBH_4 is very much less rapidly hydrolysed in water than NaAlH_4
- B) AlF_3 is insoluble in anhydrous HF but dissolves in the presence of KF
- C) If BF_3 gas is passed through Na_3AlF_6 , aluminium trifluoride is precipitated
- D) The enthalpies of dimerisation of aluminium alkyls decrease with increase in the size of alkyl groups

30. Which of the following statements is/are correct?

- A) Unlike magnesium chloride, anhydrous calcium chloride can be obtained by heating $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$
- B) Halides of elements of Group-2 with the exception of beryllium halides are mostly ionic
- C) The decrease in the solubility from BeSO_4 to BaSO_4 is primarily due to decrease in the hydration energy as one moves from Be^{2+} to Ba^{2+}
- D) The increase in the solubility of oxides and hydroxides of alkaline earth metals is primarily due to the decrease in lattice energy from Be to Ba salts

31. The correct statements among the following is/are
- A) Ionisation energy of aluminium is more than the gallium [Atomic no.31]
 - B) The ionization energy of nitrogen is more than that of oxygen
 - C) Always more is the ionization energy; less is the reactivity of the element
 - D) Lower the ionization energy, more is the basic character of the oxides of the element

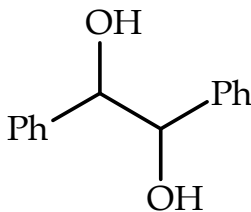
SECTION – III

(INTEGER ANSWER TYPE)

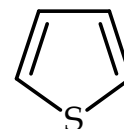
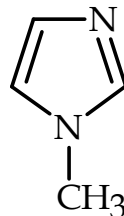
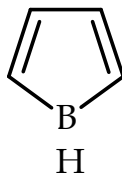
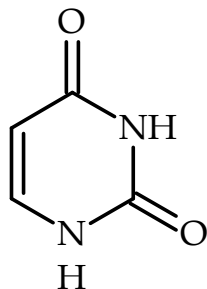
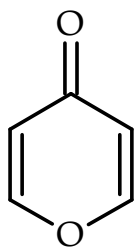
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Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases

32. A compound contains only C, H, O and Cu. In the analysis for Cu^{2+} , a 0.115 g sample of the compound is reacted with concentrated nitric acid, evaporated to dryness and the residue is dissolved in H_2O . The Cu^{2+} in the resulting solution is reacted with excess I^- and the resulting Iodine is titrated with $0.0320 \text{ M } \text{S}_2\text{O}_3^{2-}$, requiring 11.75 mL. The combustion of a 0.250 g sample of the compound produces 0.504 g of CO_2 and 0.0743 g of H_2O . Number of oxygen atoms per copper atom is... (At. Mass of Cu = 63.5 g mol^{-1})
33. Critical temperature of helium if critical pressure is 2.12 atm and critical density is 69 Kg/m^3 is..... (Assume gas follows Vanderwaal's equation of state and report answer to closest integer).
34. The number of stereo isomers possible for the following compound is



35. Among the following, the number of aromatic compound(s) is _____



36. The number of hypervalent species among the following are:



MATHEMATICS

Max Marks: 62

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

37. Let $f(x) = ax^2 + bx + c$, $a \neq 0$, $a, b, c \in I$. Suppose that $f(1) = 0$, $50 < f(7) < 60$ and $70 < f(8) < 80$, then the least values of $f(x)$ is
- A) $\frac{3}{4}$ B) $\frac{9}{2}$ C) $-\frac{9}{8}$ D) $-\frac{3}{4}$
38. An infinite sequence of digits is obtained by writing the natural numbers in order 123456789101112.....
- Define $f(n) = m$, if the 10^n th digit forms part of an m digit number. eg. $f(2) = 2$ because 100^{th} digit is 55 & it's a two digit number. The value of $f(1987)$ is
- A) 1987 B) 1984 C) 1985 D) 1986
39. A line passes through (2,0). Then which of the following is the slope of the line, for which its intercept between $y = x - 1$ and $y = -x + 1$ subtends a right angle at the origin
- A) $-\frac{2}{\sqrt{3}}$ B) $-\sqrt{3}$ C) $\sqrt{3}$ D) $\frac{1}{\sqrt{3}}$
40. A clock's minute hand has length 4 and its hour hand has length 3. The distance between the tip of the two hands at the moment when the said distance is increasing most rapidly is
- A) 5 B) 3 C) $\sqrt{7}$ D) $\sqrt{3}$

41. Let $a_0 = \frac{5}{2}$ and $a_k = a_{k-1}^2 - 2 \forall k \in N$. The value of $\prod_{k=0}^{\infty} \left(1 - \frac{1}{a_k}\right) = \frac{p}{q}$, where p & q have no common factor. Then $q - p$ is
- A) 1 B) 2 C) 3 D) 4

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

42. A solution of the trigonometric equation $\sin^{10}\left(\theta - \frac{\pi}{4}\right) + \cos^{10}\left(\theta - \frac{\pi}{4}\right) = 1$ can lie in $(\theta \in (0, 2\pi))$
- A) 1st quadrant B) 2nd Quadrant
C) 3rd Quadrant D) 4th Quadrant
43. Let $x, y > 0$ and $x + y = 2$ then
- A) $xy \leq 1$ B) $5x^2y < 6$
C) $27x^2y \in (0, 30]$ D) $x^3y^3(x^3 + y^3) \in (0, 2]$
44. Let $g(x)$ is twice differentiable function satisfying $g(0) = 0, g(1) = 1$. Then, which of the following is/are correct.
- A) there exist distinct $C_1, C_2 \in (0, 1)$ such that $g'(C_1) + g'(C_2) = 2$
B) there will be atleast one C such that $g'(C) = 1$ for $C \in (0, 1)$
C) there will be atleast one $C \in (0, 1)$ such that $g(C) = C$
D) if $g(\alpha) = \alpha$ for some $\alpha \in (0, 1)$ then there will exist atleast one $C \in (0, 1)$ such that $g'(C) = 0$
45. The vertices of ΔABC are $A(2, 0, 2), B(-1, 1, 1)$ and $C(1, -2, 4)$. The points D and E divide the sides AB and CA in the ratio 1 : 2 respectively. Another point F is taken in space such that perpendicular drawn from F to the plane of ΔABC , meets the plane at the point of intersection of the segment CD & BE . If the distance of F from the plane of ΔABC is $\sqrt{2}$.
- A) The volume of tetrahedron $ABCF$ is $\frac{7}{3}$ sq.unit

- B) The volume of tetrahedron $ABCF$ is $\frac{7}{6}$ sq.unit
- C) One of the equations of line AF is $\vec{r} = (2\vec{i} + 2\vec{k}) + \lambda(2\vec{k} - \vec{i})$, $(\lambda \in R)$
- D) One of the equations of line AF is $\vec{r} = (2\vec{i} + 2\vec{k}) + \mu(\vec{i} + 7\vec{k})$, $(\mu \in R)$
46. A is a matrix of order 3×3 and $A = [a_{ij}]_{3 \times 3}$. If $a_{ij} + a_{jk} + a_{ki} = 0$ holds for all $1 \leq i, j, k \leq 3$ then which of the following is/are true?
- A) A is non-singular B) A is singular
- C) $\sum_{1 \leq i, j \leq 3} a_{ij} = 0$ D) A is symmetric
47. Let $a, b, c > 0$ and $a + b + c = 6$, then $\frac{(ab+1)^2}{b^2} + \frac{(bc+1)^2}{c^2} + \frac{(ca+1)^2}{a^2}$ may be equal to
- A) $\frac{75}{4}$ B) 35 C) 15 D) 10
48. A rectangle $HOMF$ has sides $HO = 11$, $OM = 5$. A triangle ABC has H as orthocenter, O as circumcentre, M as mid-point of BC and F as foot of altitude from A , then
- A) Perimeter of $\triangle ABC$ is greater than 70
- B) Area of $\triangle ABC$ is integral
- C) One side of $\triangle ABC$ is rational
- D) All sides of $\triangle ABC$ are less than 30
49. Let $f(x) = 2 - |x - 3|$, $1 \leq x \leq 5$ and for rest of the values $f(x)$ can be obtained by using the relation $f(5x) = \alpha f(x) \forall x \in R$, the value of $f(x)$ in $[5^4, 5^5]$ for $\alpha = 2$ can be
- A) 16 B) 32 C) 64 D) 8

SECTION – III

(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases

50. Let ABC be a right angled triangle with $C = 90^\circ$, $a = 3$, $b = 4$, $c = 5$. Let I_1, I_2, I_3 are the excentres opposite to A, B, C respectively. D, E, F are the feet of perpendicular drawn from I_1 to BC , I_2 to CA , I_3 to AB respectively. If area of $\triangle DEF$ is $\frac{p}{q}$, where p and q are co-primes, then $p - q$ is _____.
51. The value of $\frac{1}{2} \left(\frac{1}{\cos^2 10^\circ} + \frac{1}{\sin^2 20^\circ} + \frac{1}{\sin^2 40^\circ} - \frac{1}{\cos^2 45^\circ} \right)$ is

52. In a triangle ABC, AB is parallel to y – axis, BC is parallel to x – axis, centroid is at (2,1). If median through C is $x - y = 1$, then the slope of median through A is...
53. The value of $6 + \log_{\frac{3}{2}} \left(\frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \dots \right)$ is
54. The solution set of inequality $\sum_{k=1}^{70} \frac{k}{x-k} \geq \frac{5}{4}$ is a union of disjoint intervals and let S be the sum of lengths of the intervals then the number of odd factors of S is _____.

KEY SHEET

1	D	2	D	3	D	4	C	5	C
6	BD	7	AC	8	ABCD	9	BD	10	ACD
11	CD	12	ABCD	13	AD	14	2	15	9
16	9	17	9	18	5				
19	B	20	C	21	B	22	D	23	C
24	ABCD	25	ABCD	26	AB	27	BD	28	C
29	ABCD	30	ABCD	31	BD	32	4	33	4
34	3	35	4	36	6				
37	C	38	B	39	D	40	C	41	D
42	ABCD	43	ABD	44	ABC	45	AC	46	BC
47	AB	48	ABCD	49	ABD	50	1	51	6
52	4	53	4	54	4				

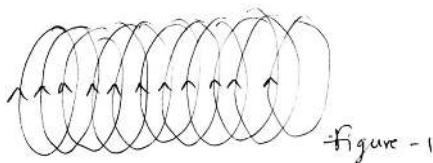

Sec: Sr. IIT
Race -15
Date: 01-03-24
PHYSICS
Max Marks: 90
SECTION-A
(TRUE/FALSE ANSWER TYPE)

This section contains 5 questions. The answer is a TRUE or FALSE.

Marking scheme +2 for correct answer , 0 if not attempted and -1 in all other cases.

1. When a point is kept at a finite distance from infinitely long conducting neutral sheet then surface charge density which would be induced is non uniform
2. Electric field at a point to any charge distribution can be calculated by gauss law.
3. In a charged non conducting solid sphere when two cavities are made randomly in it ,Then electric field must be uniform in one cavity and can be non uniform in other cavity.
4. Magnetic flux passing through the plane in which circular current carrying conductor is present, is zero.

Ans : (true)



5.

Figure-1: Stack of circular rings carrying current in the same sense

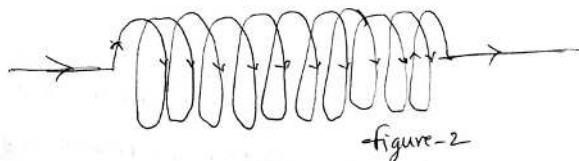


Figure-2: Infinitely long wire ,turned into solenoid in the middle (closely wounded turns).

Justify the statement that,at each and every point inside both the structures magnetic field is same

SECTION - B

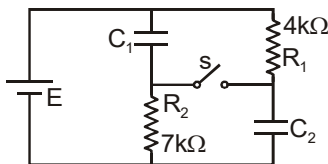
(COMPREHENSION TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS correct**.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.

Paragraph for Question Nos. 6 to 8

The switch s has been closed for long time and the electric circuit shown carries a steady current. Let $C_1 = 3.0 \mu\text{F}$, $C_2 = 6.0 \mu\text{F}$, $R_1 = 4.0 \text{ k}\Omega$, and $R_2 = 7.0 \text{ k}\Omega$. The power dissipated in R_2 is 2.8 W .



6. Choose the correct alternative (s) from the below
 - A) current in R_1 is 20 mA
 - B) current in R_1 is 2 mA
 - C) Power dissipated in R_1 is 1.6 W
 - D) power dissipated in R_1 is 2.6 W
7. Choose the correct alternative (s) from the below
 - A) the charge on capacitor C_1 is $240 \mu\text{C}$
 - B) the charge on capacitor C_1 is $840 \mu\text{C}$
 - C) the charge on capacitor C_2 is $240 \mu\text{C}$
 - D) the charge on capacitor C_2 is $840 \mu\text{C}$
8. Choose the correct alternative (s) from the below
 - A) long time after switch is opened charge on C_1 is $660 \mu\text{C}$

B) long time after switch is opened charge on C_1 is $440\mu C$

C) long time after switch is opened charge on C_2 is $880\mu C$

D) long time after switch is opened charge on C_2 is $1320\mu C$

Paragraph for Question Nos. 9 to 11

The properties of X-ray as put forward by Rontgen in his pioneering paper on the topic are as follows :

1. X-ray posses a very strong penetrative power. It can penetrate wood upto 3 cm and an aluminium foil upto 15 mm. If the hand is held between the discharge tube and the screen the dark shadow of the bones is visible within the slightly dark shadow of the hand.”
2. Photographic plates and films “show themselves susceptible to x-rays.” Hence, photography provides a valuable method of studying the effects of x-rays.
3. X-rays are neither reflected nor refracted (so far as Rontgen could discover). Hence, “X-rays cannot be concentrated by lenses.”
4. X-rays discharge electrified bodies, whether the electrification is positive or negative.
5. X-rays are generated when the cathode rays of the discharge tube strike any solid body. A heavier element, such as platinum, however, is much more efficient as a generator of x-rays than a lighter element, such as aluminum.

Most of Rontgen’s observations stood the test of time, though some of them needed to be modified later.

Now today we know if electron are accelerated through a potential difference V , then

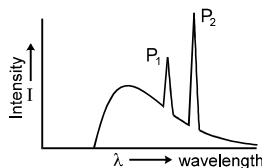
maximum energy of emitted photon could be $E_{\max} = eV$ $\frac{hc}{\lambda_{\min}} = eV, \lambda_{\min} = \frac{hc}{eV}$

$\lambda_{\min}$ is also called cut off wavelength. Since electron may loose very small energy in a given collision, the upper value of λ will approach infinity. When X-ray is produced in an X-ray tube, two types of X--ray spectra are observed : continuous spectra and line spectra.

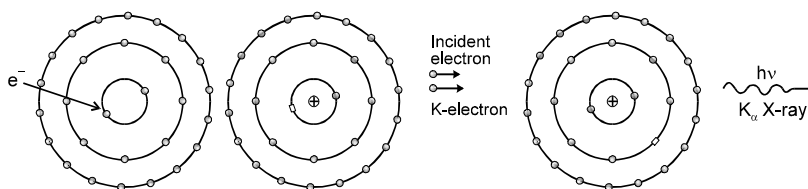
A continuous spectrum is produced by bremsstrahlung, the electromagnetic radiation produced Rontgen conducted an experiment where Parallel X-rays are passing through a optical lens available at his time with focal length 10 cm then the in correct ray diagram for the paraxial rays will be :

The electron knocks out an inner shell electron of the atom with which it collides. Let us take a hypothetical case of a target atom whose K-shell electron has been knocked out as shown.

A line spectrum results when an electron having sufficient energy collides with a heavy atom, and an electron in an inner energy level is ejected from the atom. An electron from an outer energy level then fills the vacant inner energy level, resulting in emission of an X-ray photon.



The electron knocks out an inner shell electron of the atom with which it collides. Let us take a hypothetical case of a target atom whose K-shell electron has been knocked out as shown.



This will create a vacancy in K-shell. Sensing this vacancy an electron from a higher energy state may make a transition to this vacant state. When such a transition takes place the difference of energy is converted into photon of electromagnetic radiation, which is called characteristic X-rays. Now depending upon the shell from which an electron makes a transition to K-shell we may have different lines in the K series of X-rays e.g. if electron from L shell jumps to K shell we have K_{α} , if electron from M shell jumps to K shell we have K_{β} X-ray and so on. Moseley conducted many experiments on characteristic X-rays, the findings of which played an important role in developing the concept of atomic number. Moseley's observations can be expressed as

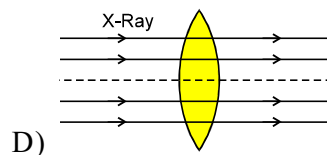
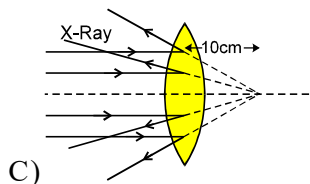
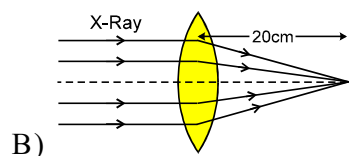
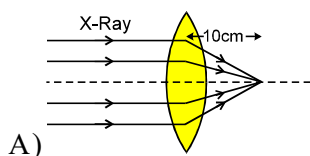
$$\sqrt{\nu} = a (Z - b)$$

where a and b are constants. Z is the atomic number of target atom and ν is the frequency.

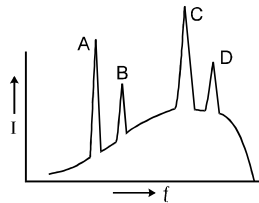
$$\sqrt{\nu} = a (Z - b)$$

where a and b are constants. Z is the atomic number of target atom and ν is the frequency.

9. In intensity v/s wavelength graph, which of the following process is responsible for the intensity of peaks P_1 and P_2 ?
- (A) Bremsstrahlung
 (B) Absorption of X-ray photons resulting in electronic excitations in atom
 (C) Emission of X-ray photons as a result of electronic transitions in atoms
 (D) Emission of x-ray photons due to acceleration of electron under influence of nucleus.
10. Rontgen conducted an experiment where Parallel X-rays are passing through a optical lens available at his time with focal length 10 cm then the in correct ray diagram for the paraxial rays will be :



11. Correct match for the graph between intensity and frequency for X-rays will be :



A) $A \rightarrow K_{\alpha}$

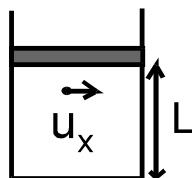
B) $B \rightarrow K_{\beta}$

C) $C \rightarrow L_{\alpha}$

D) $D \rightarrow L_{\beta}$

Paragraph for Question Nos. 12 to 14

Suppose there are N molecules each of mass m , of an ideal gas in a container. The x component of velocity of a molecule is denoted by u_x . And rms velocity of molecule is V . The gas is enclosed using a horizontal piston of area A as shown.



12. The pressure of the gas is -

(A) $\frac{m\langle u_x^2 \rangle N}{AL}$

(B) $\frac{m\langle u_x^2 \rangle N}{3AL}$

(C) $\frac{3mV^2 N}{AL}$

(D) $\frac{mV^2 N}{3AL}$

13. The pressure of the gas can also be written in terms of momentum transferred per collision(Δp) and collision frequency(f) on the wall of area A as

(A) $\Delta p \times f$ (B) $\frac{\Delta p}{A \times f}$ (C) $\frac{\Delta p \times f}{A}$ (D) $\Delta p \times f \times A$

14. If the temperature of gas is doubled keeping volume constant, we know from the gas law that the pressure will be doubled. On microscopic level this increase in pressure is because -

(A) momentum change per collision is doubled while the frequency of collision remains constant

(B) momentum change per collision remains constant while the frequency of collision is doubled

(C) momentum change per collision and the frequency of collision both are increased

(D) none of these two physical quantities are changed. It is due to some other reason.

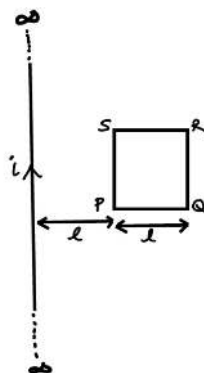
SECTION – C

(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

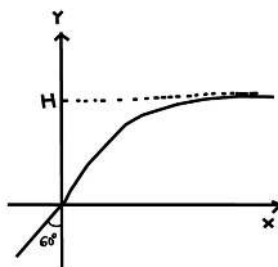
Marking scheme: +5 for correct answer, 0 if not attempted and -3 in all other cases.

15. In a hypothetical atom potential between electron and proton at distance r is given by $\left(\frac{-ke^2}{4r^3}\right)$, where k is constant. Suppose Bohr theory of atomic structure is valid and n is principle quantum number, then total energy E is proportional to
- A) n^5 B) n^7 C) n^6 D) n^4
16. A square loop PQRS of side ' ℓ ' is placed at a distance ' ℓ ' from an infinite current carrying wire as shown. Both are in the same plane. Now the loop is rotated about side PS by an angle 90° , if the resistance of loop is R , then the total amount of charge which passes through any point of the loop during rotation is



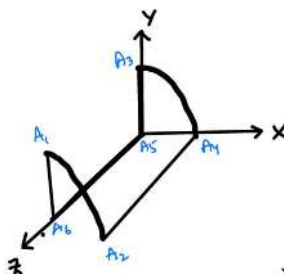
- A) $\frac{\mu_0 i l}{2\pi R} \ln \sqrt{2}$ B) $\frac{\mu_0 i l}{2\pi R} \ln 2$ C) $\frac{\mu_0 i l}{2\pi R} \ln \sqrt{3}$ D) $\frac{\mu_0 i l}{2\pi R} \ln 2\sqrt{3}$

17. A ray of light is incident at the origin at an angle 60° with y – axis as shown in figure. The refractive index is a function of y according to the relation $\mu = \frac{2}{1+y^2}$. What is the value of H shown in figure



- A) $\sqrt{\frac{2}{\sqrt{3}}}-1$ B) $\sqrt{\frac{2}{\sqrt{3}}}$ C) $\sqrt{(\sqrt{3}-1)}$ D) none

18. A time varying magnetic field $B = B_0 t \hat{k}$ is confined in a cylindrical region, cutting the XY plane on a circle of radius $x^2 + y^2 = 4$. We have placed a wire frame as shown .segment $A_1 A_2$ and $A_3 A_4$ are identical quarter circles. The net emf induced in the wire frame is equal to



- A) zero B) $2B_0$ C) $4B_0$ D) B_0

SECTION - D

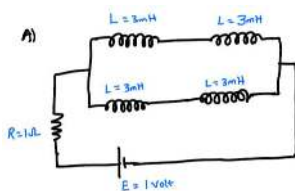
(Matching List Type)

Section-D (Matrix-Match Type, Total Marks: 24) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I, Four statements (p, q, r, s) in Column II and Four statements (w, x, y, z) in Column III. Any given statement in Column I can have correct matching with Only ONE statement given in Column II & Column III. For example, if for a given question, statement B matches with the statements given in q and x, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 3 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and -1 marks otherwise. Thus, each question in this section carries a maximum of 12 marks.

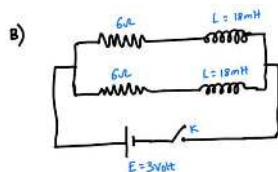
- 19 Resistance of connecting wire is negligible and L is any inductor shown in figure. Switch is closed at $t = 0$. Then match the column (Neglect mutual inductance)

Column – I

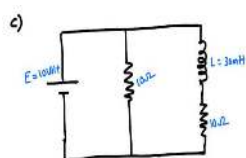
Column – II

(P) current through battery at $t = 3$ millise is

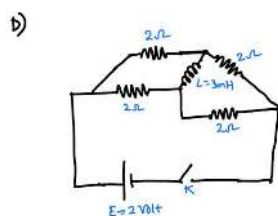
$$\left(\frac{e-1}{e}\right) \text{ amp}$$

(q) At $t = 2$ milli sec, current through

$$\text{inductor L is } \frac{1}{2} \left(\frac{e-1}{e}\right) \text{ amp}$$

(r) At $t = 3$ milli sec, current through inductor

$$\text{L is } \left(\frac{e-1}{e}\right) \text{ amp}$$



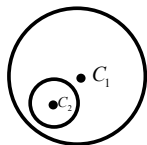
(s) Just after closing the switch current through

Inductor L is 0

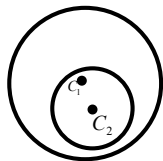
(t) Current through the battery after
very long time is 1 amp

20. **Column – I**

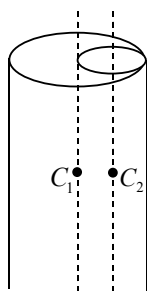
- a) Uniform positively charged
Sphere having a spherical cavity



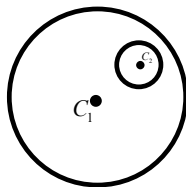
- b) Spherical cavity in a uniform
spherical Planet



- c) Cylindrical cavity in a wire carrying
uniform current density



- d) Two uniformly charged non
conducting shells outer shell is
positively charged inner shell is
negatively charged



Column – II

- p) Field at C_1 is from C_2 to C_1

- q) Field at C_1 is from C_1 to C_2

- r) Field at C_2 is from C_2 to C_1

- s) Field at C_2 is perpendicular to C_1C_2

	P	Q	R	S
(A)	3	3	3	4
(C)	4	3	4	4

	P	Q	R	S
(B)	2	1	4	2
(D)	2	3	4	3

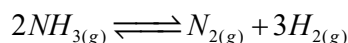
SECTION-A

(TRUE/FALSE ANSWER TYPE)

This section contains 5 questions. The answer is a TRUE or FALSE.

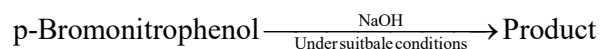
Marking scheme +2 for correct answer, 0 if not attempted and -1 in all other cases.

21. The complex $[\text{Co(en)}_3]\text{Cl}_3$ does not exhibit geometrical isomerism, but it is optically active.
22. The magnitude of heat of atomisation of metal equal to the magnitude of their heat of sublimation.
23. Ammonia under a pressure of 15 atm at 27°C is heated to 347°C in a closed vessel in the presence of a catalyst. Under the conditions, NH_3 is partially decomposed according to the equation.



The vessel is such that the volume remains effectively constant, whereas pressure increases to 50 atm. The percentage of NH_3 actually decomposed is around 61.3%.

24. Order of nucleophilicity of halide ions ($\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$) in polar protic and polar aprotic solvents is same.
25. In the following nucleophilic aromatic substitution reaction, reaction intermediate is carbanion.



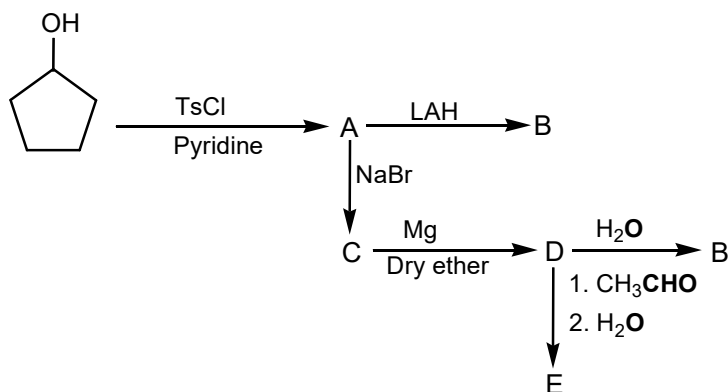
SECTION - B

(COMPREHENSION TYPE)

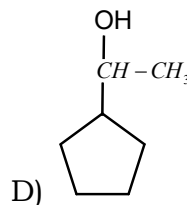
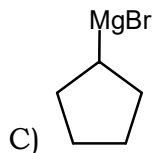
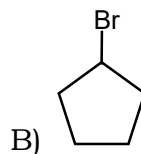
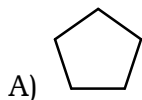
This section contains **3 Paragraph of questions**. Each paragraph has 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS correct**.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.

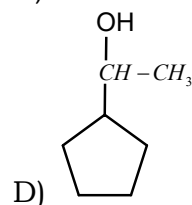
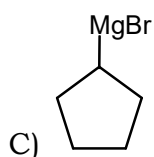
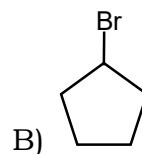
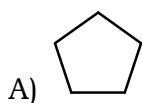
Paragraph for Question Nos. 26 to 28



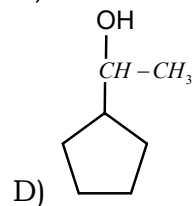
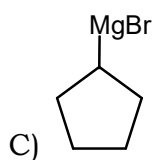
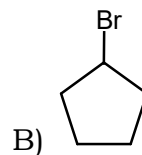
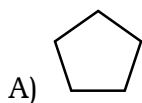
26. Compound 'C' is:



27. Compound 'D' is:



28. Compound 'E' is:



Paragraph for Question Nos. 29 to 31

Copper has the *fcc* crystal structure. Assuming an atomic radius of 130 pm for copper atom. (Cu atomic weight 63.54)

29. What is the length of unit cell of Cu?

A) 367.64 pm

B) 183.82 pm

C) 367.64 nm

D) 18.382 pm

30. How many atoms belong to the unit cell?
 A) 4 B) 2 C) 1 D) 8
31. The density of Cu is
 A) 8.54 gm/cc B) 4.27gm/cc C) 0.427 gm/cc D) 2.135gm/cc

Paragraph for Question Nos. 32 to 34

One mole of an ideal gas is initially present at 500K and $16\sqrt{2}$ atm. The gas is expanded until the final pressure fall to 1.0 atm and the final temperature dropped to 250K. Since work is a non-state function a path must be defined in order to evaluate the work done by the gas in going from initial to final state. Now, let a reversible path defined which is a combination of initial isothermal expansion followed by a subsequent adiabatic expansion such that the final state is reached. Heat capacity ($C_{v,m}$) of the gas is $1.5R$ (use : $\ln 2 = 0.7$)

32. What was the intermediate gas pressure when the process turned from isothermal to adiabatic
 A) $2.3\sqrt{2}$ atm B) $4\sqrt{2}$ atm C) $8\sqrt{2}$ atm D) $12\sqrt{2}$ atm
33. Considering both pathways i.e path I: isothermal reversible expansion followed by adiabatic reversible expansion and path II: adiabatic reversible expansion followed by isothermal reversible expansion, which of the following statements is the true regarding the net entropy change (ΔS) of the system
 A) ΔS is positive and different for both path I and path II
 B) ΔS is $1.4 R$ for both path I and path II
 C) ΔS is negative and different for both path-I & path II
 D) ΔS is $4R$ for both path I and path II.
34. For the reaction at 300 K

$$A(g) + B(g) \longrightarrow C(g)$$

$$\Delta E = -3.0 \text{ k.Cal} \quad ; \quad \Delta S = -10.0 \text{ Cal/K} \quad \text{value of } \Delta G \text{ is}$$
 A) -600 Cal B) -6600 Cal C) -6000 Cal D) $+6000 \text{ Cal}$

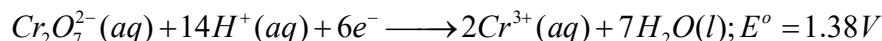
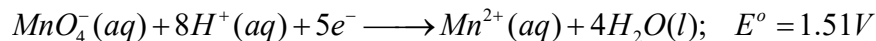
SECTION – C

(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

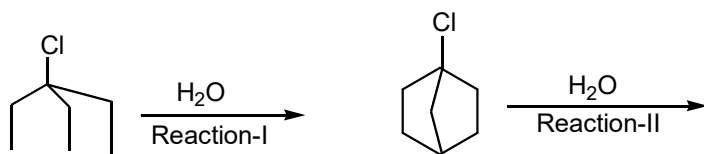
Marking scheme: +5 for correct answer, 0 if not attempted and -3 in all other cases.

35. Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration. Some half-cell reactions and their standard potentials are given below:



Identify the only *incorrect* statement regarding the quantitative estimation of aqueous $\text{Fe}(\text{NO}_3)_2$.

- A) can be used in aqueous HCl.
 B) can be used in aqueous HCl.
 C) can be used in aqueous H_2SO_4 .
 D) can be used in aqueous H_2SO_4 .
36. Phenol associates in benzene to a certain extent to form a dimer. A solution containing 20×10^{-3} kg phenol in 1 kg of benzene has its freezing point depressed by 0.69 K. Calculate the fraction of phenol that has dimerised. K_f for benzene = $5.12 \text{ kg mol}^{-1}\text{K}$.
- A) 0.27 B) 0.73 C) 0.63 D) 0.37
37. Reaction-I and reaction-II are respectively



- A) SN_1 , SN_1 B) SN_2 , SN_1 C) SN_1 , no reaction D) SN_2 , no reaction
38. The melting point of KCl is higher than that of AgCl, though crystal radii of Ag^+ and K^+ ions almost same.
- A) AgCl is more covalent than KCl, due to real inert gas configuration of Ag^+ .
 B) AgCl is more covalent than KCl, due to pseudo inert gas configuration of Ag^+ .
 C) AgCl is more ionic than KCl, due to real inert gas configuration of Ag^+ .
 D) AgCl is more ionic than KCl, due to pseudo inert gas configuration of Ag^+ .

SECTION - D

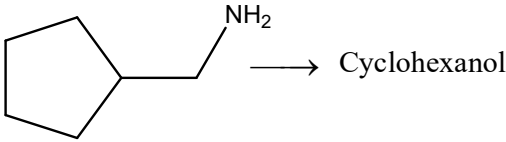
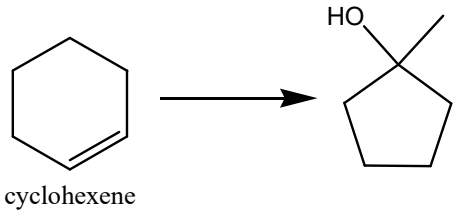
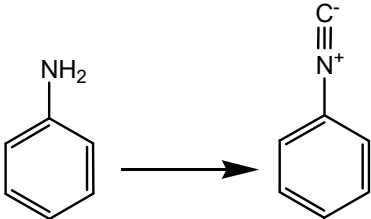
(Matching List Type)

Section-D (Matrix-Match Type, Total Marks: 24) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I, Four statements (p, q, r, s) in Column II and Four statements (w, x, y, z) in Column III. Any given statement in Column I can have correct matching with Only ONE statement given in Column II & Column III. For example, if for a given question, statement B matches with the statements given in q and x, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 3 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and -1 marks otherwise. Thus, each question in this section carries a maximum of 12 marks.

39.

COLUMN-I(MOLECULE)		COLUMN-II(Hybridisation of central atom)		COLUMN-III (Number of lone pair)	
a)	OSF_4	P)	sp^3d^2	W)	0
b)	$TeCl_4$	Q)	sp^3	X)	2
c)	XeO_6^{4-}	R)	sp^3d	Y)	1
d)	S_3O_9 (SO_3 trimer)	S)	sp^2	Z)	3

40.

COLUMN-I(Reaction)		COLUMN-II(Reagent)		COLUMN-III(intermediate)	
a)	$RCH_2COOH \longrightarrow R(CH_2)_2COOH$	P)	H_3O^+	W)	Carbocation
b)		Q)	$NaNO_2/HCl$	X)	Ketene
c)	 cyclohexene	R)	Diazomethane	Y)	Dichlorocarbene
d)		S)	$CHCl_3/NaOH$	Z)	Carbanion

SECTION-A

(TRUE/FALSE ANSWER TYPE)

This section contains 5 questions. The answer is a TRUE or FALSE.

Marking scheme +2 for correct answer , 0 if not attempted and -1 in all other cases.

41. The equation of plane about which the image of the origin containing the plane $x + 2y - z = 0$ lies on the plane containing the lines $\frac{x-2}{4} = \frac{y-2}{-1} = \frac{z-7}{2}$ and $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-4}{4}$ is $2x + 4y - 2z + 3 = 0$
42. The tangent at point P on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through the point $(0, -b)$ and the normal at point P passes through the point $(2\sqrt{2}a, 0)$. If e denote the eccentricity of hyperbola then the value of e^2 is 4
43. Sum of all positive real solutions of the equation $2 \cos 2x \left(\cos 2x - \cos \frac{2014\pi^2}{x} \right) = \cos 4x - 1$ is $K\pi$ then, sum of digits of K is 9
44. Let W, K and S denote the events associated with one day cricket match of India against Australia as W: India wins the match, K: Kohli hits a century, C: Shami takes 5 wickets. The following probabilities are known as $P\left(\frac{W}{K \cap S}\right) = 0.4, P\left(\frac{W}{\overline{K} \cap S}\right) = 0.3, P\left(\frac{W}{K \cap \overline{S}}\right) = 0.2, P\left(\frac{W}{\overline{K} \cap \overline{S}}\right) = 0.1$. If India bat first then $P(K) = 0.3, P(S) = 0.3$ and if India bat second then $P(K) = 0.5, P(S) = 0.2$. Assume that team which wins the toss, will bat first. If India wins the match then probability that Kohli hit a century and Shami took 5 wickets is $\frac{\lambda}{\mu}$ where $\lambda, \mu \in N$. The least value of $\mu - \lambda$ is 4
45. If Z is a non-zero complex number such that $\left| Z^3 + \frac{1}{Z^3} \right| \leq 2$, then the maximum value of $\left| Z + \frac{1}{Z} \right|$ is 4

SECTION - B

(COMPREHENSION TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS correct**.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.

Paragraph for Question Nos. 46 to 48

Let $f(x)$ be a twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(2-x)$ and

$$f'\left(\frac{1}{2}\right) = f'\left(\frac{1}{4}\right) = 0. \text{ Then}$$

46. The minimum number of values where $f''(x)$ vanishes on $[0, 2]$ is

- A) More than 2 B) less than 8 C) 4 D) 5

47. $\int_{-1}^1 f'(1+x)x^2 e^{x^2} dx =$

- A) an integer B) 2 C) an irrational number D) 0

48. $\int_0^1 f(1-t)e^{-\cos \pi t} dt - \int_1^2 f(2-t)e^{\cos \pi t} dt =$

- A) $\int_0^2 f'(t)e^{\cos \pi t} dt$ B) 0 C) 2 D) $\int_{2\pi}^{4\pi} \frac{\sin^{-1}(\sin x)}{\sin^4 x + \cos^2 x} dx$

Paragraph for Question Nos. 49 to 51

$A(-1, -1); B(1, -1); C(1, 1)$ and $D(-1, 1)$ are the vertices of a square ABCD. C_1 is a circle inscribed in it and C_2 is a circle circumscribing the square. A circle C touches the circle C_1 externally and also a fixed line $L = 0$ in its plane

49. Which of the following statements is/are correct

- A) Locus of centre of C is an ellipse
 B) Locus of centre of C is a parabola
 C) If $L \equiv 3x + 4y - 20 = 0$ and p is any arbitrary point on C_1 then the least distance of p from L is 3
 D) If $L \equiv 3x + 4y - 20 = 0$ then, the smallest radius of C is $3/2$

50. A variable parabola passes through the points where C_1 cuts X-axis and has an arbitrary tangent of C_2 as its directrix. The locus of the focus of the variable parabola is S then

A) Equation of S is $\frac{x^2}{2} + \frac{y^2}{1} = 1$

B) equation of S is $\frac{x^2}{1} + \frac{y^2}{2} = 1$

C) equation of director circle of S is $x^2 + y^2 = 3$

D) foci of S are $(0, \pm 1)$

51. A line M through A is drawn parallel to BD. Point S moves such that its distance from the line BD and the vertex A are equal. If locus of S cuts M at T_2 and T_3 and AC at T_1 , then area of $\Delta T_1 T_2 T_3$ is

A) less than 2 sq units

B) $\frac{2}{3}$ sq units

C) 1 sq unit

D) more than 1 sq unit

Paragraph for Question Nos. 52 to 54

Consider two straight lines $L_1 : -2x + 4 = 2y - 4 = z - 4$ and $L_2 : 2x = y + 1 = -z + 3$ where the points P and Q lies on L_1 and L_2 respectively and point R divides PQ in the ratio 2:3 (internally). Also d is the distance between the points P and Q

52. The value of d if PQ is parallel to Z-axis is

A) 3

B) 4

C) 5

D) less than 6

53. The locus of R is

A) $2x + z = 6$

B) $y + 2z + 2 = 0$

C) contains the point $(3, -2, 0)$

D) $2x + 3y + z = 0$

54. The equation of plane containing the point P when PQ is parallel to Z-axis and perpendicular to the line of intersection of the planes $3x + 4y + 7z + 4 = 0$ and $x - y + 2z + 3 = 0$,

A) is $45x + 3y - 21z + 16 = 0$

B) is $x + y + 3z - 18 = 0$

C) contains the point $\left(1, 1, \frac{64}{21}\right)$ D) is $3x + 3y + 6z - 40 = 0$ **SECTION – C****(SINGLE CORRECT ANSWER TYPE)**

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +5 for correct answer, 0 if not attempted and -3 in all other cases.

55. Let $f(x) = (\cos^2 x) e^{\tan x}$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then which of the following is incorrect

A) $f'(x)$ has a point of local minima at $x = \frac{\pi}{4}$

B) $f'(x)$ has a point of local maxima in $\left(-\frac{\pi}{4}, 0\right)$

C) $f'(x)$ has exactly two points of local maxima/ minima in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

D) $f''(x) = 0$ has no root in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

56. If the value of definite integral $\int_{\frac{1}{3}}^{\frac{1}{2}} \left\{ x \left[\frac{2}{x} \right] \right\} dx = \frac{m}{n}$ where m and n are coprime then the absolute value of $(m - n)$ is

[Note: $[k]$ denotes greatest integer less than or equal to k and $\{k\}$ denotes fractional part of k

A) 375

B) 368

C) 389

D) 412

57. The largest integral value of 'a' for which every solution of the equation, $x([x] - 5) + 2\{x\} + 6 = 0$ satisfies the inequality $(a - 3)x^2 + 2(a + 3)x - 8a \leq 0$, where $[y]$ and $\{y\}$ denote greatest integer and fractional part function respectively is

A) 0

B) 1

C) 2

D) 3

58. Let $f(x) = \begin{cases} x^2 + 1 & x < 0 \\ (x-1)^3 + 2b, & x \geq 0 \end{cases}$ and $g(x) = \begin{cases} x + 2a & \text{if } x < 0 \\ |x-2|, & \text{if } x \geq 0 \end{cases}$

(Where $a \in \mathbb{R}^-$ and $b \in \mathbb{R}$). If $\text{fog}(x)$ is continuous for all real x and differentiable at $x=0$, then find the value of $8(b-a)$ is

A) 18

B) 17

C) 15

D) 21

SECTION - D

(Matching List Type)

Section-D (Matrix-Match Type, Total Marks: 24) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I, Four statements (p, q, r, s) in Column II and Four statements (w, x, y, z) in Column III. Any given statement in Column I can have correct matching with Only ONE statement given in Column II & Column III. For example, if for a given question, statement B matches with the statements given in q and x, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 3 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and -1 marks otherwise. Thus, each question in this section carries a maximum of 12 marks.

59.

	COLUMN-I		COLUMN-II		COLUMN-II
a)	The number of integral values of k for which $e^x = \frac{k}{x-3}$ has exactly two solutions is m and the maximum integral value of k for which $e^x = \frac{k}{x-3}$ has no solution is $-M$	P)	$m=2$	W)	$M=5$
b)	$\lim_{n \rightarrow \infty} \frac{e\left(1 - \frac{1}{n}\right)^n - 1}{n^\alpha}$ exists finitely and is equal to a non zero constant C. Let $m = 12(C - \alpha)$ and $M =$	Q)	$m=7$	X)	$M=6$

c)	Consider the function $f(x) = \frac{\sqrt{x^2+1}}{x-1}$ let m denote the number of integers which are not in the range of $f(x)$ and M denote the total number of local minima; local maxima and asymptotes of $f(x)$	R)		Y)	
			$m=6$		$M=8$
d)	The parabola $y^2 = x - y$ divides the triangle formed by the tangent to it at origin and the straight lines $x + y = 0$ and $y = -1$ into two areas α and β where $\alpha < \beta$ given $m = 24\alpha$ and $M = 12\beta$	S)		Z)	
			$m=8$		$M=3$

60.

	COLUMN-I		COLUMN-II		COLUMN-II
a)	The circle $C_1 : x^2 + y^2 = 3$ with centre at O intersects the parabola $x^2 = 2y$ at the point p in 1 st quadrant let the tangent to the circle C_1 at p touch two other circles C_2 and C_3 at R_2 and R_3	P)		W)	
			$m=9$		$M=2$

	respectively, suppose C_2 and C_3 have equal radii $2\sqrt{3}$ and centers Q_2 and Q_3 respectively and Q_2, Q_3 lie on y-axis let $m = Q_2Q_3$ and $M = \frac{3.(R_2R_3)^2}{(Ar \ O R_2R_3)^2}$				
b)	Three boxes are labeled as x, y, z and each box containing 5 balls numbered 1, 2, 3, 4, 5. The balls in each box are well mixed and one ball is choosen at random from each of three boxes x, y, z. α, β, γ are the numbers on the balls respectively choosen and p is the probability that $\alpha = \beta + \gamma$ given , $m = 50P$ $M = 125P$	Q)	m=4	X)	M=30
c)	Let X be a 3×3 matrix satisfying $X \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$ $X \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \text{ and}$	R)	m=12	Y)	M=4

* * * * *

Key SheetPHYSICS

1-10	True	False	Fals	True	Fals	AC	AC	AD	AC	ABC
11-20	AB	AC	C	C	C	A	A	A	(A) $\rightarrow$ (p,q,s,t), (B) $\rightarrow$ (p,q,s,t) (C) $\rightarrow$ (r,s), (D) $\rightarrow$ (s,t)	a-q; b-p; c-s; d-q

CHEMISTRY

21-30	True	True	True	False	True	B	C	D	A	A
31-40	A	B	B	A	A	C	C	B	a-R-W; b-R-Y; c-P-W; d-Q-W	a-R-X; b-Q-W; c-P-X; d-S-Y

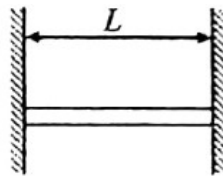
MATHEMATICS

41-50	Fal	Fals	True	True	False	ABC	AD	ABD	BCD	AC
51-60	A	CD	AC	AC	D	C	B	C	a) $a \rightarrow Q; Y$ b) $b \rightarrow R; Y$ c) $c \rightarrow P; Z$ d) $d \rightarrow S; Y$	a) $a \rightarrow R; Y$ b) $b \rightarrow Q; Z$ c) $c \rightarrow P; X$ d) $d \rightarrow S; W$


Sec: Sr. IIT
Race -16
Date: 02-03-24
PHYSICS
Max Marks : 60
Section-1
(One or more options correct type)

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D).
 Out of Which Only One is correct

1. A horizontal rod (length L , mass m , cross section A , coefficient of linear expansion α) fits exactly (with out tension) between two rigid vertical supports choose the correct option



A) Since the length of rod cannot increase beyond L , on increasing temperature, the effect of thermal expansion is neutralized by compressive forces developed by the wall

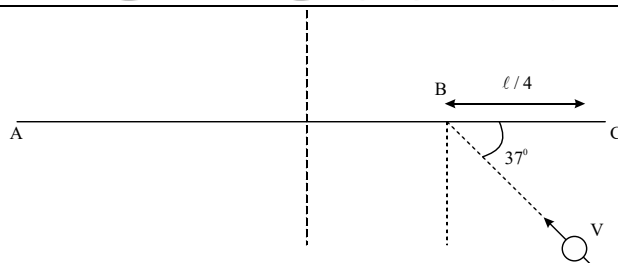
B) If the walls and rod are rough with mutual friction coefficient μ , then increment in temperature θ of the rod, required to prevent the rod from falling is

$$\Delta\theta = \frac{mg}{2\mu YA\alpha}$$

C) If the walls and rod are rough with mutual friction coefficient μ , then increment in temperature θ of the rod, required to prevent the rod from falling is $\Delta\theta = \frac{mg}{4\mu YA\alpha}$

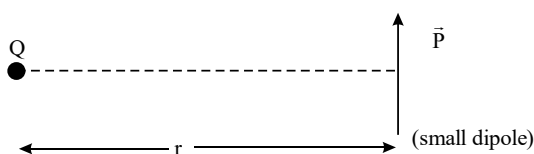
D) The stress developed in wire is $\alpha y \Delta\theta$

2. A uniform rod AC of length l and mass m is kept on a horizontal smooth plane. It is free to rotate and move. A particle of same mass m moving on the plane with velocity v strike the rod at point B making an angle 37° with the rod. The collision is elastic. After collision.



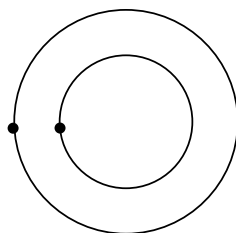
- A) The angular velocity of the rod will be $\frac{72v}{55l}$
- B) The centre of the rod will travel a distance $\frac{\pi l}{3}$ in the time in which it makes half rotation
- C) Impulse of the impact force is $\frac{24mv}{55}$
- D) All the above are correct

3. For the situation shown in the figure (Assume $r \gg$ length of dipole) mark out the correct statements.



- A) Force acting on the dipole is zero
- B) Force acting on the dipole is $\frac{PQ}{4\pi\epsilon_0 r^3}$
- C) Torque acting on the dipole is $\frac{2PQ}{4\pi\epsilon_0 r^2}$
- D) Torque acting on the dipole is $\frac{PQ}{2\pi\epsilon_0 r^2}$

- 4) Two concentric, coplanar, circular loop of wire, with different diameter carry current in the same sense as shown in the figure. Which of the following statement is correct?

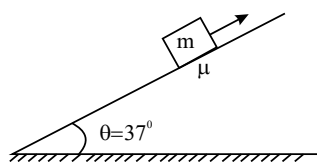


- A) The magnetic force exerted by the outer loop on a short portion of the inner loop is radially outward
- B) The magnetic force exerted by the outer loop on a short portion of the inner is radially outward

- C) The net magnetic force exerted by the outer loop on a whole inner is non zero and is radially outward
- D) The net magnetic force exerted by the outer loop on a whole inner loop is zero

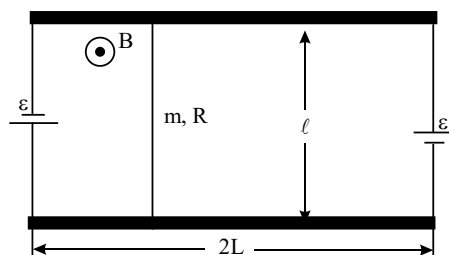
- 5). A block of mass m is being pulled up the rough incline, inclined at an angle 37° with horizontal by an agent delivering constant power P . The coefficient of friction between the block and the incline is μ . Find the maximum speed (in m/s) of the block during the course of ascent.

[Take : $P = 60\text{W}$, $m = 1\text{kg}$, $\mu = 0.5$]



- A) 1 B) 3 C) 5 D) 6
6. Side rail of length $2L$ are fixed on a horizontal plane at a distance l from each other. These ends are connected by two identical ideal batteries with emf E by resistanceless wires (see figure). On the rails is a rod of mass m , which may slide along them. The entire system is placed in a uniform vertical magnetic field B .

Assuming that the resistance of the rod is R and the resistance per unit length of each of the rails equal to ρ , find the period of small oscillations (in sec.) arising from shifting the rod from the equilibrium along the rails. Neglect friction, internal resistance of batteries and induced emf in the rod. [Take : $B = \pi T$, $\varepsilon = \pi$ volt, $l = 0.5m$, $L = 1m$, $\rho = 1\Omega/m$, $R = 0.25\Omega$, $m = 100\text{gm}$]

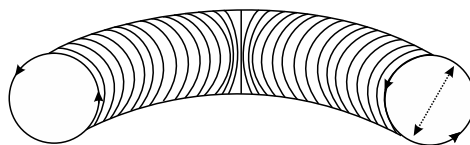


- A) 1 B) 3 C) 5 D) 6

7. A wooden plank of length 1m and uniform cross section is hinged at one end to the bottom of a tank as shown in the figure. The tank is filled with water up to a height of 0.5m. The specific gravity of the plank is 0.5. θ is angle which plank makes with the vertical in the equilibrium position (exclude the case $\theta = 0$). Find the value of $\frac{1}{\cos^2 \theta}$.

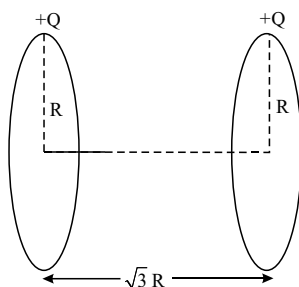
A) 1 B) 2 C) 3 D) 4

8. Calculate the magnetic moment (in Am^2) of a thin wire with a current $I = 8.0\text{A}$, wound tightly on a half a toroid (see figure). The diameter of the cross section of the toroid is equal to $d = 5.0\text{cm}$, and the number of turns is $N = 500$.



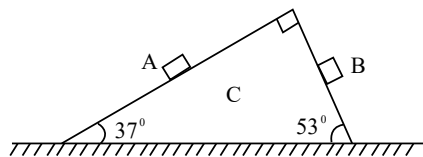
A) 5 B) 6 C) 7 D) 8

9. Two identical charged rings of charge $Q=1\text{C}$ and radius $R = 0.1\text{m}$ are separated by a distance $\sqrt{3}R$ meter, and are fixed. A point charge of charge $q = 1.0 \times 10^{-9}\text{C}$ and mass $m = 2\text{kg}$ is moved by external agent from the centre of left ring to the centre of right ring. Find the work done by an external agent in joule.



A) 0 B) 1 C) 2 D) 3

- 10). In the figure shown blocks 'A' and 'B' are kept on a wedge 'C'. A, B and C have mass m . All surfaces are smooth. Find the acceleration of C.



A) 0 B) 1 C) 2 D) 3

SECTION - II

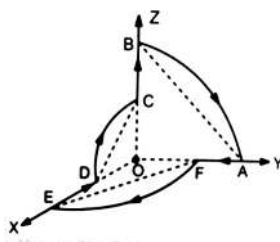
(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 11 & 12

In a rigid wire frame ABCDEFA, as shown in the figure, a current I_0 flows through the wire from A to F as shown in the figure. Radii of the three quarters AB, CD and EF are $r_1 = 3^{\frac{1}{4}}m$, $r_2 = 3^{\frac{-1}{4}}m$ and $r_3 = 1m$ respectively with common centre at O. All the wires are light except for the portion ED which have a mass of 'm' kg.



11. The magnetic moment of the loop is

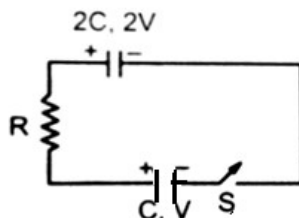
- A) $\frac{\pi I_0}{4} \left[\sqrt{3}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \hat{k} \right]$ B) $-\frac{\pi I_0}{4} \left[\sqrt{3}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \hat{k} \right]$
 C) $\frac{\pi I_0}{4} \left[-\sqrt{3}\hat{i} - \frac{1}{\sqrt{3}}\hat{j} + \hat{k} \right]$ D) $\frac{\pi I_0}{4} \left[-\sqrt{3}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \hat{k} \right]$

12. If this loop is kept in a magnetic field of magnetic 2T which is directed along positive y-axis, angular acceleration of the loop is: (where l =length of portion ED)

- A) $\frac{8\pi I_0}{ml^2} [\hat{i} + \sqrt{3}\hat{k}]$ B) $\frac{6\pi I_0}{ml^2} [\hat{i} + \sqrt{3}\hat{k}]$ C) $\frac{8\pi I_0}{ml^2} [\hat{i} - \sqrt{3}\hat{k}]$ D) $\frac{6\pi I_0}{ml^2} [\hat{i} - \sqrt{3}\hat{k}]$

Paragraph for Question Nos. 13 & 14

Two capacitors of capacitance $2C$ and C are charged to potential difference $2V$ and V respectively. Now the charged capacitors are connected with a resistor R as shown in figure. Switch 'S' is closed at $t=0$.



13. What is the total heat produced in circuit till steady state is obtained?

- A) $\frac{1}{3}CV^2$ B) $\frac{CV^2}{2}$ C) $\frac{2}{3}CV^2$ D) $\frac{3}{5}CV^2$

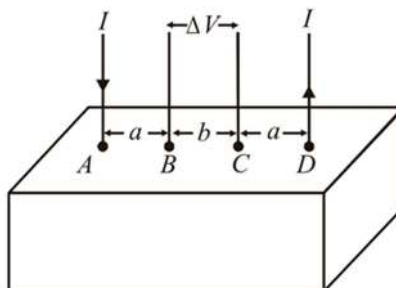
14. What is the current in circuit as function of time?

- A) $\frac{V}{R}e^{-3t/RC}$ B) $\frac{2V}{R}e^{-2t/3RC}$ C) $\frac{V}{3R}e^{-t/2RC}$ D) $\frac{V}{R}e^{-3t/2RC}$

Paragraph for Question Nos. 15 & 16

Consider a block of conducting material of resistivity ρ shown in the figure. Current I enters at A and leaves from D. We apply superposition principle to find voltage ΔV developed between B and C. The calculation is done in the following steps:

- Take current I entering from A and assume it to spread over a hemispherical surface in the block.
- Calculate field $E(r)$ at distance r from A by using Ohm's law $E = \rho j$, where j is the current per unit area at r .
- From the r dependence of $E(r)$, obtain the potential $V(r)$ at r .
- Repeat (i), (ii) and (iii) for current I leaving D and superpose results for A and D.



15. ΔV measured between B and C is

- A) $\frac{\rho I}{\pi a} - \frac{\rho I}{\pi(a+b)}$ B) $\frac{\rho I}{a} - \frac{\rho I}{(a+b)}$ C) $\frac{\rho I}{2\pi a} - \frac{\rho I}{2\pi(a+b)}$ D) $\frac{\rho I}{2\pi(a-b)}$

16. For current entering at A, the electric field at a distance ' r ' from A is

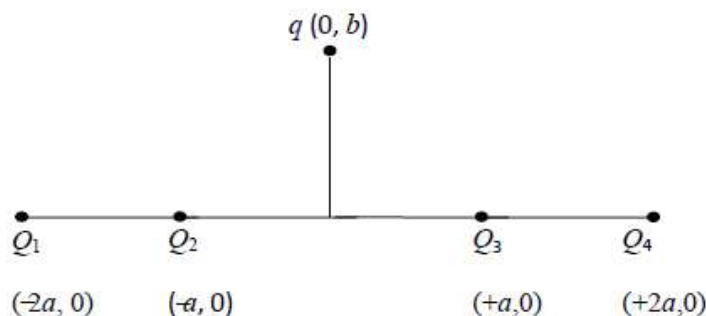
- A) $\frac{\rho I}{8\pi r^2}$ B) $\frac{\rho I}{\pi r^2}$ C) $\frac{\rho I}{2\pi r^2}$ D) $\frac{\rho I}{4\pi r^2}$

Section-3

(Matching List Type)

This section contains four questions, each having two matching lists (**List-1** & **List-II**). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x axis $x = -2a, -a, +a$ and $+2a$, respectively. A positive charge q is placed on the positive y axis at a distance $b > 0$. Four options of the signs of these charges are given in List I. The direction of the forces on the charge q is given in List II. Match List I with List II and select the correct answer using the code given below the lists.



LIST-I		LIST-II	
P	Q_1, Q_2, Q_3, Q_4 all positive	1	+x
Q	Q_1, Q_2 positive: Q_3, Q_4 negative	2	-x
R	Q_1, Q_4 positive: Q_2, Q_3 negative	3	+y
S	Q_1, Q_3 positive: Q_2, Q_4 negative	4	-y

A) P-3, Q-1, R-4, S-2

B) P-4, Q-2, R-3, S-1

C) P-3, Q-1, R-2, S-4

D) P-4, Q-2, R-1, S-3

18. Four combinations of two thin lenses are given in List I. The radius of curvature of all curved surfaces is r and the refractive index on all the lenses is 1.5. Match lens combinations in List I with their focal length in List II and select the correct answer using the code given below the lists.

List I

List II

P.

1. $2r$

Q.

2. $r/2$

R.

3. $-r$

S.

4. r **Choices :**

A) P-1, Q-2, R-3, S-4

B) P-2, Q-4, R-3, S-1

C) P-4, Q-1, R-2, S-3

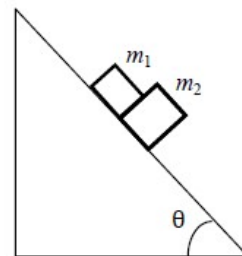
D) P-2, Q-1, R-3, S-4

19. A block of mass $m_1 = 1\text{ kg}$ another mass $m_2 = 2\text{ kg}$, are placed together (see figure) on an inclined plane with angle of inclination θ . Various values of θ are given in List I. The coefficient of friction between the block m_1 and the plane is always zero. The coefficient of static and dynamic friction between the block m_2 and the plane are equal to $\mu = 0.3$. In List II expressions for the friction on block m_2 are given. Match the correct expression of the friction in List II with the angles given in List I, and choose the correct option. The acceleration due to gravity is denoted by g .

[Useful information : $\tan(55^\circ) \approx 0.1$; $\tan(11.5^\circ) \approx 0.2$; $\tan(16.5^\circ) \approx 0.3$]

List I

List II

P. $\theta = 5^\circ$ 1. $m_2 g \sin \theta$ Q. $\theta = 10^\circ$ 2. $(m_1 + m_2) g \sin \theta$ R. $\theta = 15^\circ$ 3. $\mu m_2 g \cos \theta$ S. $\theta = 20^\circ$ 4. $\mu(m_1 + m_2) g \cos \theta$ 

Choices :

- A) P-1, Q-1, R-1, S-3 B) P-2, Q-2, R-2, S-3
C) P-2, Q-2, R-2, S-4 D) P-2, Q-2, R-3, S-3

20. A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance of 1.2 m from the person. In the following state of the lift's motion is given in List I and the distance where the water jet hits the floor of the lift is given List II. Match the statements from List I with those in List II and select the correct answer using the code given below the list

List I

- P. List is accelerating vertically up
Q. Lift is accelerating vertically down with an acceleration less than the gravitational acceleration
R. Lift moving vertically up with constant speed
S. List is falling freely

List II

1. $d = 1.2m$
2. $d > 1.2m$
3. $d < 1.2m$
4. No water leaks out of the jar

- A) P-2, Q-3, R-2, S-4 B) P-2, Q-3, R-1, S-4
C) P-1, Q-1, R-1, S-4 D) P-2, Q-3, R-1, S-1

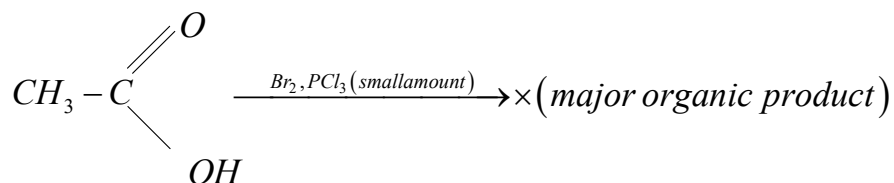
Section-1

(One or more options correct type)

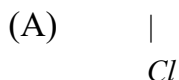
This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D).

Out of Which Only One is correct

21.



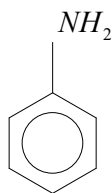
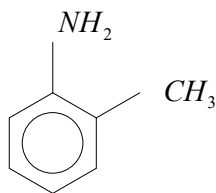
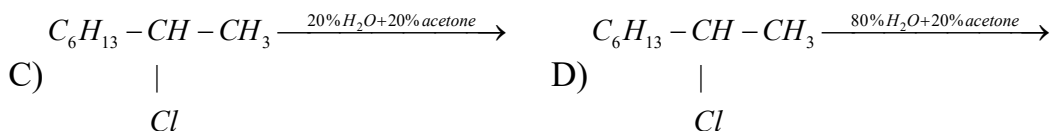
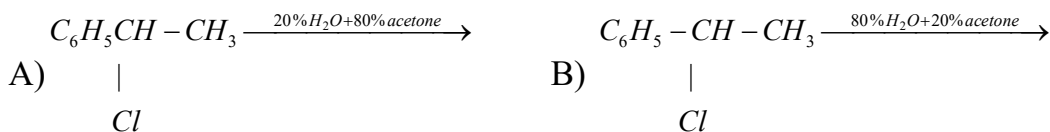
Incorrect statement among the following is

'X' is $\text{CH}_2 - \text{COOH}$ (B) In the formation of 'X' CH_3COCl intermediate involved

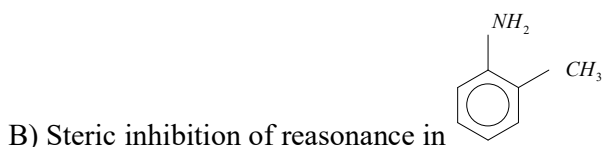
(C) In the formation of 'X' enolate intermediate involved

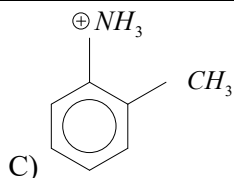
(D) $\text{Br}_2 + \text{Red P}_4$ can also be used to get 'X'

22. Among the following reaction in which one percentage of racemization is highest

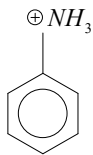


23. is less basic than in aqueous solution the main reason for it is

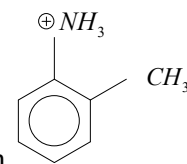
A) Here $-\text{CH}_3$ acts as electron withdrawing group



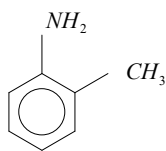
is less stable than



, due to steric inhibition of hydration in



D) Due to crowding from $-CH_3$,



can not accept proton easily when compared with aniline

24. Heat of combustion of n-Hexane, n-heptane and n-octane are -1002.4 k cal/mol, -1159.9 k cal /mol and -1317.2 k cal/mol respectively. If heat of combustion of 2- methylpentane is -990 k cal/mole, heat of combustion of 2- methylnonane is nearly
- A) -1618 k.cal/mol B) -1650 k cal/mol
C) -1550 k cal/mole D) -1700 k cal/mole
25. Absolute value of which of the following thermodynamic property of a substance can be calculated?
- A) Internal energy B) Enthalpy
C) Gibbs energy D) Entropy
26. Liquid A and liquid B are immiscible liquids. At $25^{\circ}C$ vapour pressures of pure liquid A and pure liquid B are 100 mm of Hg and 200 mm of Hg respectively. Total vapour pressure of mixture of 2 moles liquid A and 3 moles of liquid B at $25^{\circ}C$ is
- A) 160 mm of Hg B) 200 mm of Hg C) 250 mm of Hg D) 300 mm of Hg
27. At 298 k the standard free energy of formation of $H_2O_{(l)}$ is -237.23 kJ/mole, standard free energy change for ionization $H_2O_{(l)}$ to $H^+_{(aq)}$ and $OH^-_{(aq)}$ is +79.71 kJ/mole. The reversible EMF at 298 k of the cell is $Pt; H_2(1bar) | H^+(1M) || OH^-(1M) | O_2(1bar); Pt$
- A) +0.403V B) +0.806 V C) +0.2 V D) +1.0V
28. EAN of iron in $Fe_2(CO)_9$ is
- A) 34 B) 35 C) 36 D) 37

29. When molten ICl subjected to electrolysis with inert electrodes, anode product is/are
- A) I_2 only B) Cl_2 only C) Both I_2 and Cl_2 D) ICl_3
30. Among the following halides for which one ΔH_f° more negative
- A) LiF B) NaF C) RbF D) CsF

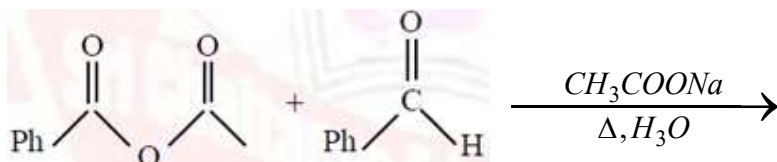
Section-2

(Paragraph Type)

This section contains 3 paragraphs each describing theory, experiment, data etc. Six questions relate to three paragraphs with two questions on each paragraph. Each question pertaining to a particular paragraph should have **only one correct answer** among the four choices A, B, C and D.

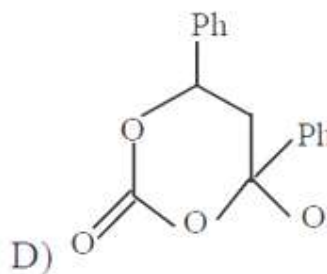
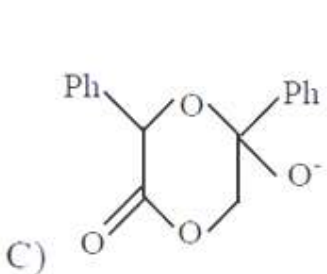
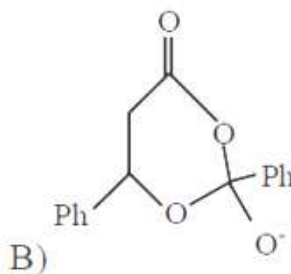
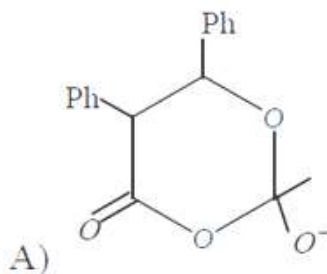
Paragraph for Questions 31 & 32

Condensation reaction between aromatic aldehyde and aliphatic acid anhydride in presence of sodium or potassium salt of the acid corresponding to the anhydride yields α, β - unsaturated aromatic acid

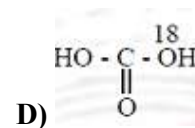
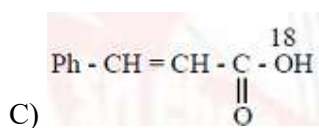
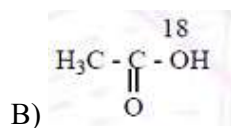
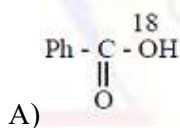


(Observe the above reaction)

31. The intermediate product in the above reaction is



32. If $\text{Ph}-\text{CH}=\text{O}^{18}$ is used instead of $\text{PhCH}=\text{O}$ in original reaction the " O^{18} " presence shall be observed in which product formed (assuming released water molecule in course of reaction involves in hydrolysis in priority at final step)



Paragraph for Questions 33 & 34

Imagine an atom made up of a proton and a hypothetical particle of half the mass of the electron but having the same charge as the electron. Bohr's model is applicable and all transition of this hypothetical particle from higher level to ground state is possible.

33. The largest wavelength of photon that will be emitted when hypothetical particle jumps to first excited level is
- (A) $\frac{18}{5R}$ (B) $\frac{72}{5R}$ (C) $\frac{36}{5R}$ (D) $\frac{8}{R}$
34. The velocity of this particle in second Bohr orbit is
- (A) $2.188 \times 10^6 \text{ m/sec}$ (B) $1.094 \times 10^6 \text{ m/sec}$ (C) $0.547 \times 10^6 \text{ m/sec}$ (D) $4.376 \times 10^6 \text{ m/sec}$

Paragraph for Questions 35 & 36

A chemist opened a cupboard to find four bottles containing water solutions, each of which has lost its label. Bottle 1, 2 and 3 contained colourless solutions, while bottle 4 contained a blue solution. The labels from the bottles were lying scattered on the floor of the cupboard. They were:

Copper (II) sulphate, Hydrochloric acid, Lead nitrate Sodium carbonate

By mixing samples of the contents of the bottles, in pairs, the chemist made the following observation:

Bottle 1 + Bottle 2	White precipitate	Bottle 1 + Bottle 3	White precipitate
Bottle 1 + Bottle 4	White precipitate	Bottle 2 + Bottle 3	Colourless gas evolved
Bottle 2 + Bottle 4	No visible reaction	Bottle 3 + Bottle 4	Blue precipitate

35. Bottle 3 contains
- A) copper (II) sulphate B) hydrochloric acid
C) lead nitrate D) sodium carbonate
36. When bottle 1 is mixed with bottle 4, white precipitate is observed, which is
- A) PbSO_4 B) CuCO_3 C) PbCl_2 D) CuCl_2

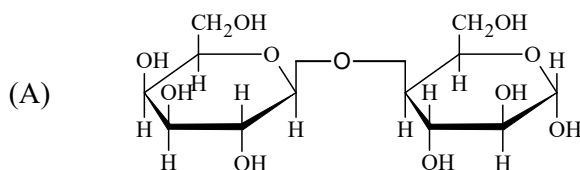
Section-3

(Matching List Type)

This section contains four questions, each having two matching lists (**List-1** & **List-II**). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

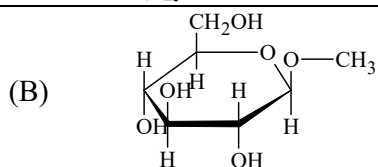
37. Match the Column:

Column – I



Column – II

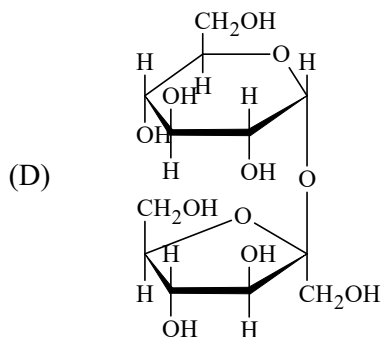
(p) glycosidic linkage



(q) reducing sugar

(C) Starch

(r) oligosaccharide



(s) non-sugar

The correct match is

(A) A-PQR,B-PQR,C-PS,D-P

(B) A-PQ,B-PQR,C-PS,D-PR

(C) A-PQR,B-PQ,C-P,D-PR

(D) A-PQR,B-P,C-PS,D-PR

38. **Column – I****Column – II****(Arrangement in unit cell, radius ratio in higher limit)****(Coordination number of cation : Anion)**

A) Cations in CCP and anions in alternate tetrahedral voids

P) ratio of number of cation to anion in one unit cell is 1 : 1

B) Cations in simple cubic and anions in the body centre.

Q) ratio of coordination number of cation to anion is 1 : 1

C) Anions in CCP and cations in all tetrahedral voids.

R) number of next neighbours of ion is greater than 10.

D) Cations in CCP and anions in all octahedral voids. S) Effective number of formula units = 4

The correct match is

A) A – P, Q,S,; B – P,Q,R,S; C – P,S,; D – P,Q,S,

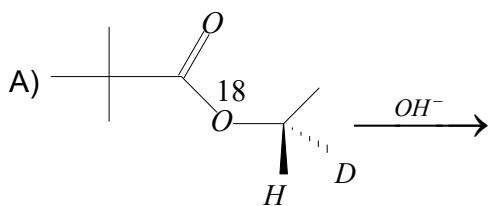
B) A – P,Q; B – P,Q,R,S; C – S,; D – P,Q,S,

C) A – P,Q,R,S,; B – P,Q,; C – R,S; D – P,Q,R,S

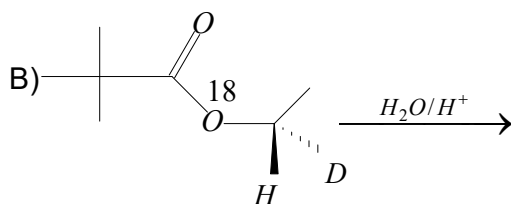
D) A – P,Q,S,; B – P,Q,R; C – P,Q,S,; D – S

39. Match the column:

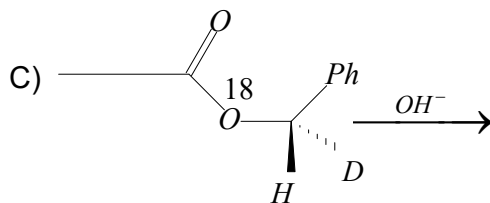
Column – I



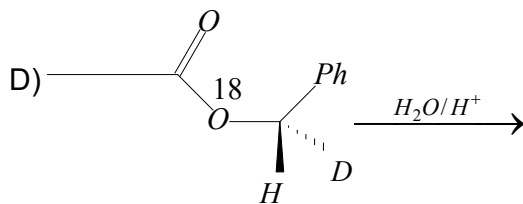
P) Bimolecular path



Q) Racemic mixture of alcohol



R) Acyl oxygen cleavage

S) Alcohol with O¹⁸

T) Inversion (only) of configuration in alcohol part

The correct match is

A) A-P,R,S; B-P,R,S; C-Q,T; D-QT

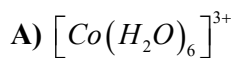
B) A-R,S ; B-P,T; C-Q ; D-P,R

C) A-P,T; B- R,S; C-P,R,S; D-Q

D) A-P,S; B-R,S; C-Q; D-P,R

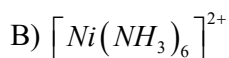
40. Match the column:

Column – I

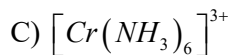


Column – II

P) Low spin complex



Q) Paramagnetic complex



R) Inner orbital complex



S) Two bonds (metal-ligand) are longer than the remaining four bonds

The correct match is

A) A-P,R; B-P,Q,R; C-P,Q,R; D-QS

B) A-P,R; B-Q; C-Q,R; D-Q,S

C) A-Q,S; B-Q; C-Q,R; D-Q,S

D) A-Q,S; B-P,R,S; C-Q,R; D-Q

MATHEMATICS

Max Marks : 60

Section-1

(Only one Option correct Type)

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D). Out of Which Only One is correct

41. If 6, 8 and 12 are $l^{\text{th}}, m^{\text{th}}$ and n^{th} terms of an A.P. and $f(x) = nx^2 + 2lx - 2m$, then the equation

$f(x) = 0$ has

A) a root between 0 and 1

B) both roots imaginary

C) both roots negative

D) both roots greater

42. If $f(x) = \frac{1}{2} + \int_0^x (t-x) \sin t \, dt$ for $x \in [0, \pi]$ if M and m are the maximum and minimum values of

$f(x)$ respectively in $[0, \pi]$ then $(M - m)$ equals

A) $\frac{1}{2}$

B) $\frac{\pi}{2}$

C) π

D) 3π

43. The ellipse $E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$ is Inscribed in a rectangle R whose sides are parallel to the co-ordinate axes. Another ellipse E_2 passing through the point $(0, 4)$ circumscribes the rectangle R. The eccentricity of the ellipse E_2 is

A) $\frac{1}{\sqrt{2}}$

B) $\frac{\sqrt{3}}{2}$

C) $\frac{1}{2}$

D) $\frac{3}{4}$

44. If $1, \alpha_1, \alpha_2, \dots, \alpha_{2009}$ are $(2009)^{\text{th}}$ roots of unity, then the value of $2008 \sum_{r=1}^{2008} r(\alpha_r + \alpha_{2009-r})$ equals :

A) 2009

B) 2008

C) 0

D) -2009

45. There are 'n' urns $u_1, u_2, u_3 \dots u_n$. Each urn contains $(2n+1)$ balls. The i^{th} urn contains $2i$ number of white balls. Let $p(ui)$ which is probability of selecting i^{th} urn is proportional to $i^2 + 2$. If we randomly select one of the urns and draw one ball and probability of ball being white be $P(A)$; then $\lim_{n \rightarrow \infty} P(A)$ is equal to
- A) $\frac{3}{4}$ B) $\frac{1}{4}$ C) $\frac{2}{3}$ D) $\frac{3}{16}$
46. If $\sin(\sin x + \cos x) = \cos(\cos x - \sin x)$, then the value of $\sin x$ can be
- A) $\frac{\pi}{6}$ B) $\frac{\sqrt{16 - \pi^2}}{4}$ C) $\frac{\pi}{8}$ D) $\frac{-\pi}{6}$
47. If the distance from the point $P(1,1,1)$ to the line passing through the points $Q(0,6,8)$ and $R(-1,4,7)$ is expressed in the form $\sqrt{\frac{p}{q}}$ where p, q are co-prime, then the value of $\frac{(p+q)(p+q-1)}{2}$ equals
- A) 4950 B) 5050 C) 5150 D) 5250
48. Which one of the following statements does not hold good for the function $f(x) = \cos^{-1}(2x^2 - 1)$?
- A) f is not differentiable at $x = 0$
- B) f is monotonic
- C) f is even
- D) f has an Extremum
49. If a circle of radius 3 units is touching the lines $\sqrt{3}y^2 - 4xy + \sqrt{3}x^2 = 0$ in the first quadrant, then the length of the chord of contact to the circle is
- A) $\frac{\sqrt{3}+1}{2}$ B) $\frac{\sqrt{3}+1}{\sqrt{2}}$ C) $3\left(\frac{\sqrt{3}+1}{\sqrt{2}}\right)$ D) $3\left(\frac{\sqrt{3}+1}{2}\right)$

50. If $2a + 2b + 3c = \frac{1}{5}$ and $a, b, c > 0$ then maximum value of term of Independent of x in the Expansion

of $\left(abx^{\frac{1}{2}} + cx^{-\frac{1}{3}} \right)^{25}$ is

- A) $25_{C_{10}}$ B) $25_{C_{15}} (35)^{25}$ C) $25_{C_{15}} \left(\frac{1}{35} \right)^{35}$ D) $25_{C_{10}} \left(\frac{1}{25} \right)^{35}$

SECTION - II

(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS correct**. **Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

Paragraph for Question Nos. 51 to 52

Suppose $f(x)$ and $g(x)$ are two continuous functions defined for $0 \leq x \leq 2$. Given

$$f(x) = \int_0^1 e^{x+t} f(t) dt \text{ and } g(x) = \int_0^1 e^{x+t} g(t) dt + x$$

51. The value of $g(0) - f(0)$ equals.

- A) $\frac{2}{3-e^2}$ B) $\frac{3}{e^2-2}$ C) $\frac{1}{e^2-1}$ D) 0

52. The value of $\frac{g(0)}{g(2)}$ equals :

- A) 0 B) $\frac{1}{3}$ C) $\frac{1}{e^2}$ D) $\frac{2}{e^2}$

Paragraph for Question Nos. 53 to 54

A bag contains 4 white, 3 black and 2 red balls. Balls are drawn from the bag one by one without replacement.

53. The probability that 4th ball is red is equal to

- A) $\frac{5}{36}$ B) $\frac{1}{12}$ C) $\frac{2}{9}$ D) $\frac{3}{11}$

54. The probability that 5th ball is black given that 4th ball is red is equal to

- A) $\frac{1}{8}$ B) $\frac{3}{8}$ C) $\frac{2}{11}$ D) $\frac{3}{11}$

Paragraph for Question Nos. 55 to 56

Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the parabola $y^2 = 2x$. They intersect at P and Q in the first and fourth quadrants respectively. Tangents to the ellipse at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

55. The ratio of the areas of the triangles PQS and PQR, is

- A) 1 : 3 B) 1 : 2 C) 2 : 3 D) 3 : 4

56. The equation of circle touching the parabola at upper end of its latusrectum and passing through its vertex, is

- A) $2x^2 + 2y^2 - x - 2y = 0$ B) $2x^2 + 2y^2 + 4x - \frac{9}{2}y = 0$
 C) $2x^2 + 2y^2 + x - 3y = 0$ D) $2x^2 + 2y^2 - 7x + y = 0$

SECTION - III**(Matching List Type)**

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

57.

	Column-I		Column-II
P)	Let $f(x) = \cos\left(\tan^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)$ Then $f\left(\frac{1}{4}\right)$ is equal to	1)	7
Q)	A bag contains 4 white, 3 black and 2 red balls. Balls are drawn from the bag one by one without replacements. If probability of 4 th ball is red equals $\frac{p}{q}$ where p and q are relatively prime, then $(q-p)$ equals	2)	4
R)	The area bounded by the curve $y = x + \sin x$ and its inverse function between the ordinates $x = 0$ to $x = 2\pi$ is 4s. Then	3)	2

	the value of s is		
S)	Let A_i where $i = 1, 2, 3, \dots, 12$ are the vertices of a regular dodecagon and G is its center. Let 's' be the number of straight lines that can be formed with these 13 points, 't' be number of triangles that can be formed and 'd' be number of diagonals in a dodecagon ; then the value of $\frac{s+t+d}{100}$	4)	1

	P	Q	R	S		P	Q	R	S
A)	4	1	2	3	B)	4	1	3	2
C)	1	4	2	3	D)	1	4	3	2

58.

	Column-I		Column-II
P)	If the coefficient of x^2 and x^3 in the expansion of $(1+8x+bx^2)(1-3x)^9$ in the power of x are equal , then $\frac{7b}{27}$ is	1)	5
Q)	The number of integral values of α for which the equation $x^2 + 2(\alpha-1)x + \alpha + 5 = 0$ has both roots positive and distinct is	2)	3
R)	Let $I = \int_0^1 (1-x^{50})^{99} x^{100} dx$ and	3)	2

	$J = \int_0^1 (1-x^{50})^{100} x^{100} dx$. If $\frac{I}{J}$ is equal to $\frac{m}{n}$ where m and n are co-prime then the value of $\frac{m-n-1}{20}$		
S)	The slopes of sides BC, CA, AB of triangle ABC whose orthocenter is origin are $-1, -2, -3$ respectively. If locus of centroid of triangle ABC is $y = \left(\frac{m}{n}\right)x$. Where m,n are relatively prime then n-m is	4)	1

	P	Q	R	S
A)	3	2	1	4
B)	3	2	4	1
C)	3	2	1	1
D)	3	2	4	4

59.

	Column-I		Column-II
P)	A tangent to the hyperbola $\frac{x^2}{4} - y^2 = 1$ meets the ellipse $x^2 + 4y^2 = 4$ at two distinct points . The locus of mid-point of this chord is $(x^2 + 4y^2)^2 = k(x^2 - 4y^2)$, then 'k' is equal to	1)	2
Q)	Let the equation of the plane containing the line $x - y - z - 4 = 0 = x + y + 2z - 4$ and is parallel to the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$ be $x + Ay + Bz + C = 0$, then the value of $ A + B + C $	2)	11
R)	Let ABC be a triangle whose centroid G, orthocenter is H and circumcentre is the origin O. If 'D' is any point in the plane of the triangle such that no three of O, A,B,C and D are collinear satisfying the relation $\overrightarrow{AD} + \overrightarrow{BD} + \overrightarrow{CH} + 3\overrightarrow{HG} = \lambda \overrightarrow{HD}$ then value of the scalar $\lambda + 4$ is	3)	4

S)	If $\alpha = \sin \left(\cot^{-1} \left(\tan \left(\cos^{-1} \left(\frac{2}{3} \right) \right) \right) \right)$ and $\beta = \sin \left(\cos ee^{-1} \left(\cot \left(\tan^{-1} \left(\frac{1}{3} \right) \right) \right) \right)$ are the roots of the quadratic equation $ax^2 + bx + c = 0$ where a,b,c are integers and 'c' is prime then the value of $(a+b+c)$ equals	4)	6
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	P	Q	R	S
A)	2	1	4	3
B)	1	2	4	3
C)	3	2	4	1
D)	3	4	1	2

60. 6 married couples are present in a birth day party number of ways in which four persons are selected if :

	Column-I		Column-II
P)	They form no pair	P)	15
Q)	They form at least one pair	Q)	240
R)	They form at most one pair	R)	255
S)	They form exactly two pair	S)	480

	P	Q	R	S		P	Q	R	S
A)	2	4	3	1	B)	2	3	1	4
C)	3	2	4	1	D)	2	3	4	1

KEY SHEET

01	C	02	D	03	B	04	A	05	D	06	A
07	D	08	C	09	A	10	A	11	B	12	C
13	A	14	D	15	A	16	C	17	A	18	B
19	D	20	C								
21	A	22	A	23	C	24	A	25	D	26	D
27	A	28	C	29	C	30	A	31	B	32	A
33	B	34	B	35	D	36	A	37	D	38	C
39	C	40	B								
41	A	42	C	43	C	44	D	45	A	46	B
47	A	48	B	49	C	50	C	51	A	52	B
53	C	54	B	55	C	56	D	57	B	58	C
59	C	60	D								